

Masters Thesis



CFD INVESTIGATION OF CARBON MONOXIDE COMBUSTION AND SCRAP PREHEATING IN SHAFT FURNACES FOR HEAT RECOVERY OPTIMIZATION

Von

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Die vorliegende Arbeit mit dem Titel

CFD INVESTIGATION OF CARBON MONOXIDE COMBUSTION AND SCRAP PREHEATING IN SHAFT FURNACES FOR HEAT RECOVERY OPTIMIZATION

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Vorwort

Die vorliegende Masterarbeit mit dem Thema „CFD Investigation of Carbon Monoxide Combustion and Scrap Preheating in Shaft Furnaces for Heat Recovery Optimization" wurde in Zusammenarbeit mit der Firma Primetals Technologies Germany GmbH durchgeführt.

Für die Betreuung und Unterstützung während der gesamten Arbeit möchte ich mich herzlich bei meinem firmeninternen Betreuer Dr. Ing. Andreas Opfermann bedanken, der mir trotz eines anspruchsvollen industriellen Umfelds stets mit fachlichem Rat und wertvollen Einblicken zur Seite stand.

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Abstract

Electric arc furnace (EAF) shaft preheating systems recover heat from hot off-gas by passing it through a packed bed of scrap metal. Under good conditions, this can save up to 100 kWh per ton of steel. The off-gas enters the shaft at around 1500 K and contains 15–20% carbon monoxide by mass. CO combustion could release an additional 283 kJ/mol of heat, but plant measurements consistently show CO escaping the shaft uncombusted. The mechanisms behind this were not well understood, and direct measurements inside an operating shaft are not practical.

This thesis develops a CFD model using ANSYS to investigate heat transfer, gas flow, and CO combustion in EAF shaft preheating systems. The scrap bed is modeled as a porous medium with porosity $\varepsilon = 0.85$. Flow resistance is calculated using the Ergun equation and gas-solid heat transfer is modeled with the Gunn correlation. Separate energy equations are solved for the gas and solid phases to capture the transient temperature difference between them. Turbulence is modeled with the Realizable k- ε model, and CO combustion is modeled using the Eddy Dissipation Model, which is appropriate for mixing-limited reactions as confirmed by Damköhler number analysis ($Da \approx 5800$). Gas properties are calculated as functions of temperature using VDI Heat Atlas correlations and Wilke's mixing rule.

Transient simulations of 450 seconds capture one preheating batch, starting from cold scrap at 300 K and hot gas at 1500 K. Parametric studies cover off-gas mass flow rates from 5 to 45 kg/s and particle diameters from 0.05 to 0.30 m. Convective heat transfer coefficients range from 36 to 119 W/(m²·K) with increasing mass flow rate, and average scrap temperatures reach approximately 325°C after 5 minutes under baseline conditions. Smaller particles heat significantly faster due to higher surface area, which explains plant observations of localized melting despite moderate predicted bulk temperatures.

Combustion simulations show that stoichiometric oxygen becomes available from air infiltration at suction pressures as low as 800 Pa. Despite this, CO still escapes the shaft, indicating that the problem is not oxygen deficiency but insufficient mixing between the CO-rich off-gas and the infiltrating air. Isolated gap configurations reveal that front and back infiltration creates combustion zones in the center of the shaft, while side infiltration provides oxygen near the walls but does not penetrate the core. Increasing air flow through larger gaps or higher suction improves combustion to some extent but also cools the scrap, resulting in a clear trade-off between combustion efficiency and preheating performance.

Kurzfassung

Vorgeheizte Elektrolichtbogenöfen (EAF) mit Schachtvorwärmssystemen nutzen die Wärme des heißen Abgases, indem sie es durch ein Schüttbett aus Schrott leiten. Unter günstigen Bedingungen können dabei bis zu 100 kWh pro Tonne Stahl eingespart werden. Das Abgas tritt mit etwa 1500 K in den Schacht ein und enthält 15 bis 20 Massenprozent Kohlenmonoxid. Die CO-Verbrennung könnte zusätzlich 283 kJ/mol Wärme freisetzen, jedoch zeigen Anlagenmessungen konsistent, dass CO den Schacht unverbrannt verlässt. Die Mechanismen dahinter waren bislang nicht vollständig verstanden, und direkte Messungen in einem in Betrieb befindlichen Schacht sind nicht praktikabel.

Diese Arbeit entwickelt ein CFD-Modell mit ANSYS, um Wärmeübertragung, Gasströmung und CO-Verbrennung in EAF-Schachtvorwärmssystemen zu untersuchen. Das Schüttbett wird als poröses Medium mit einer Porosität von $\varepsilon = 0,85$ modelliert. Der Strömungswiderstand wird mit der Ergun-Gleichung berechnet, und die Gas-Feststoff-Wärmeübertragung wird mit der Gunn-Korrelation modelliert. Für Gas- und Festphase werden getrennte Energiegleichungen gelöst, um den transienten Temperaturunterschied zwischen beiden Phasen zu erfassen. Die Turbulenz wird mit dem Realizable k- ε -Modell modelliert, und die CO-Verbrennung wird mit dem Eddy-Dissipation-Modell abgebildet, das für mischungslimitierte Reaktionen geeignet ist, wie durch die Damköhler-Zahl-Analyse bestätigt wurde ($Da \approx 5800$). Die Gaseigenschaften werden als Temperaturfunktionen mithilfe der VDI-Wärmeatlas-Korrelationen und der Mischungsregel nach Wilke berechnet.

Transiente Simulationen über 450 Sekunden erfassen einen vollständigen Vorwärmbatch, beginnend mit kaltem Schrott bei 300 K und heißem Gas bei 1500 K. Parameterstudien umfassen Abgasmassenströme von 5 bis 45 kg/s sowie Partikeldurchmesser von 0,05 bis 0,30 m. Die konvektiven Wärmeübergangskoeffizienten liegen zwischen 36 und 119 W/(m²·K) mit steigendem Massenstrom, und die mittleren Schrotttemperaturen erreichen nach 5 Minuten unter Basisbedingungen etwa 325 °C. Kleinere Partikel erwärmen sich aufgrund der größeren spezifischen Oberfläche deutlich schneller, was Anlagenbeobachtungen lokalisierter Schmelzvorgänge trotz moderater vorhergesagter Schütttemperaturen erklärt.

Verbrennungssimulationen zeigen, dass stöchiometrischer Sauerstoff durch Lufteströmung bereits bei Unterdrücken von 800 Pa verfügbar wird. Trotzdem verlässt CO den Schacht noch immer, was darauf hinweist, dass das Problem nicht ein Sauerstoffmangel ist, sondern eine unzureichende Durchmischung zwischen dem CO-reichen Abgas und der einströmenden Luft. Isolierte Spaltgeometrien zeigen, dass Lufteströmung von vorne und hinten Verbrennungszonen im Zentrum des Schachts erzeugt, während seitliche Einströmung Sauerstoff in Wandnähe bereitstellt, aber nicht in den Kern vordringt. Eine Erhöhung des Luftstroms durch größere Spalte oder höheren Unterdruck verbessert die Verbrennung in gewissem Maß, kühlt den Schrott jedoch gleichzeitig ab, woraus ein klarer Zielkonflikt zwischen Verbrennungseffizienz und Vorwärmleistung resultiert.

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Nomenklatur

Roman Symbols

A	Pre-exponential factor (Arrhenius), dimensionless
A, B, C, D, E, F, G	VDI polynomial fitting coefficients
A_{EDM}	Eddy dissipation model constant, dimensionless
A_{fs}	Interfacial area density (fluid-solid), m^2/m^3
A_{p}	Particle surface area, m^2
B_{EDM}	Eddy dissipation model product inhibition constant, dimensionless
c_{p}	Specific heat capacity, $\text{J}/(\text{kg}\cdot\text{K})$
C_{μ}	Turbulence model coefficient, dimensionless
$C_{1\varepsilon}, C_{2\varepsilon}$	k- ε turbulence model constants, dimensionless
C_2	Inertial resistance coefficient, m^{-1}
C_{d}	Discharge coefficient, dimensionless
d_{p}	Particle diameter, m
D_{i}	Molecular diffusion coefficient of species i, m^2/s
D_{eff}	Effective diffusion coefficient, m^2/s
D_{turb}	Turbulent diffusivity, m^2/s
Da	Damköhler number, dimensionless
E_{a}	Activation energy, J/kmol
F_{ij}	Wilke's mixture interaction factor, dimensionless
G_{k}	Production of turbulent kinetic energy, $\text{kg}/(\text{m}\cdot\text{s}^3)$
h_{fs}	Fluid-solid heat transfer coefficient, $\text{W}/(\text{m}^2\cdot\text{K})$
H	Bed height, m
I	Turbulence intensity, %
k	Turbulent kinetic energy, m^2/s^2
k_{g}	Gas thermal conductivity, $\text{W}/(\text{m}\cdot\text{K})$
k_{s}	Solid thermal conductivity, $\text{W}/(\text{m}\cdot\text{K})$
L	Bed height, m
L_{c}	Characteristic length, m
l_{t}	Turbulence length scale, m
$\dot{m}$	Mass flow rate, kg/s

M	Molecular weight, kg/kmol
$\dot{n}$	Molar flow rate, kmol/s
n	Number of elements/cells, dimensionless
Nu	Nusselt number, dimensionless
P	Pressure, Pa
Pe	Peclet number, dimensionless
Pr	Prandtl number, dimensionless
Q	Heat/thermal power, W or MW
R	Universal gas constant, J/(kmol·K)
R_{CO}	Volumetric reaction rate for CO, kmol/(m ³ ·s)
Re	Reynolds number, dimensionless
Re_p	Particle Reynolds number, dimensionless
s	Stoichiometric mass ratio, kg/kg
S_{ct}	Turbulent Schmidt number, dimensionless
S_i	Momentum source term, kg/(m ² ·s ²)
S_q	Energy source term, W/m ³
t	Time, s
T	Temperature, K
$\mathbf{v}$	Velocity vector, m/s
v_s	Superficial velocity, m/s
V	Volume, m ³
x_i	Mole fraction of species i, dimensionless
Y_i	Mass fraction of species i, dimensionless

Greek Symbols

α	Permeability, m ²
ΔH_c^0	Standard enthalpy of combustion, kJ/mol
ΔP	Pressure drop, Pa
ΔT	Temperature difference, K
ε	Porosity (void fraction), dimensionless
ε	Turbulent dissipation rate, m ² /s ³
λ	Thermal conductivity, W/(m·K)

μ	Dynamic viscosity, Pa·s
μ_t	Turbulent (eddy) viscosity, Pa·s
ρ	Density, kg/m ³
$\sigma_k, \sigma_\varepsilon$	Turbulence model constants, dimensionless
τ	Stress tensor, Pa
τ_{chem}	Chemical time scale, s
τ_{res}	Residence time, s
τ_{thermal}	Thermal time constant, s

Subscripts and Superscripts

avg	Average
chem	Chemical
comb	Combustion
eff	Effective
f	Fluid
final	Final state
fs	Fluid-solid
g	Gas phase
i	Species index / direction
in	Inlet
inject	Injected
inlet	Inlet condition
max	Maximum
mix	Mixture
out	Outlet
p	Particle
recovery	Energy recovery
req	Required
res	Residence
s	Solid phase / steel
scrap	Scrap
sensible	Sensible heat

solid	Solid
steel	Steel
t	Turbulent
thermal	Thermal
total	Total
turb	Turbulent
0	Initial condition / standard state

Abbreviations

CFD	Computational Fluid Dynamics
CO	Carbon Monoxide
CO ₂	Carbon Dioxide
DRI	Direct Reduced Iron
EAF	Electric Arc Furnace
EDM	Eddy Dissipation Model
H ₂ O	Water (vapor)
HTC	Heat Transfer Coefficient
N ₂	Nitrogen
O ₂	Oxygen
RANS	Reynolds-Averaged Navier-Stokes
VDI	Verein Deutscher Ingenieure (Association of German Engineers)

1. Background and Context

Steel is a material with very wide applications in industry due to its properties which can be tailored for different requirements in industry by either forming various alloys or different treatments. Due to the economic growth and increase in population, demand for steel is increasing continuously. From 1950 to 2023, steel production increased from 189 million tons to 1892 million tons, making it ten times higher. With the increase in steel demand and production environmental concerns also increase [1].

Steel production is a process that uses significant amounts of energy and heavily depends on coal. According to World Steel Association, steel production is making approximately 7-8% of global greenhouse gas emissions at around 1.9 tons of CO₂ per ton of crude steel cast, and it is responsible for about 7% of energy consumption globally, using around 20GJ of energy per ton of crude steel cast. Coal is the main energy source in steel production in blast furnaces, for slag foaming in electric arc furnaces (EAF) and recovery of iron oxides through chemical reduction inside the furnace during operation or in direct reduced iron (DRI) production [1].

Currently the blast furnaces make up the bigger portion of steel production at around 70% across the world with higher coal consumption at and higher carbon emissions 1630-1960kg CO₂ per ton of steel. Whereas DRI based EAF steel production emitting 560-1450kg CO₂ per ton of steel and scrap-based EAF steel production at around 360-470kg of CO₂ per ton of steel. Reason for higher CO₂ emissions with DRI based operations, DRI production process also using carbon for iron reduction which is now also steering towards finding alternatives such as usage of hydrogen or natural gas in the reduction process. Also, DRI melting in an EAF requires higher energy for melting between 301-640kWh per ton of steel and the source of electrical energy needs to be considered for indirect carbon emissions. While many regions are shifting toward EAF with scrap and DRI based production, blast furnaces remain essential for meeting global steel demand and currently it is not feasible to quit blast furnaces all together due to insufficient availability of quality scrap, DRI and high-grade iron ore [2,3].

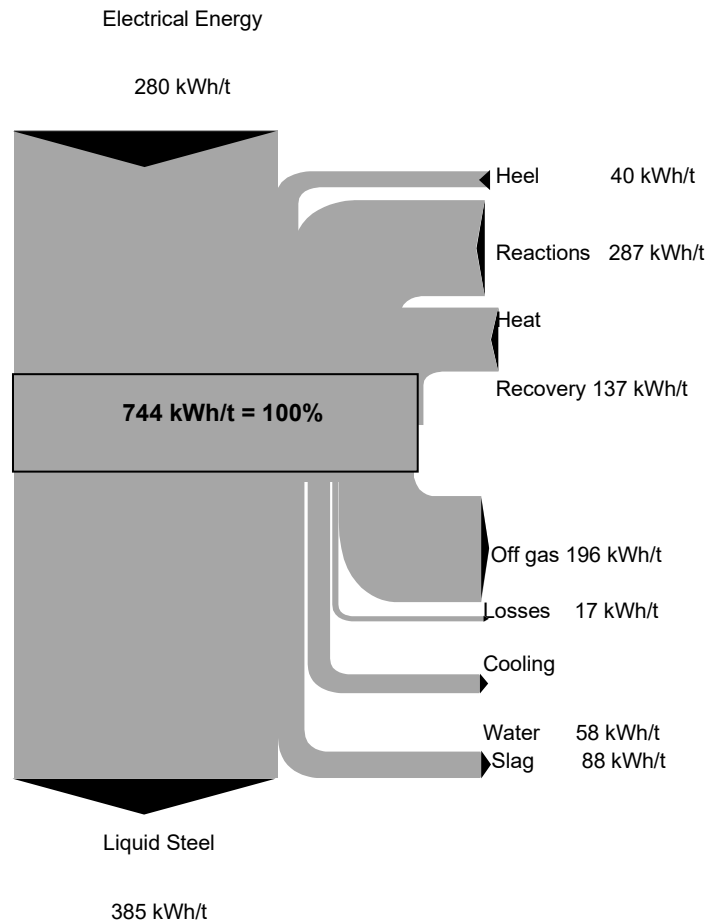


Figure 1.1 EAF Energy Balance

Electric Arc Furnaces represents a more sustainable alternative to traditional blast furnace production methods, using recycled steel scrap and DRI, consuming 40-60% less energy per ton of steel produced. As the steel industry faces increasing pressure to reduce carbon emissions and improve energy efficiency, optimization of steel making processes has become critical.

Steel industry is focusing on increasing energy efficiency and decreasing environmental impacts of steel making processes to meet the requirements of global climate goals. To do so, research and development advances to find improvements in areas such as increasing system efficiency by internal heat recovery and one scrap preheating method is investigated in this study using CFD.

Problem Statement

Electric Arc Furnace shaft preheating systems use hot off gas coming from the furnace at around 1500K containing 15-20% CO by passing it through packed scrap beds to recover thermal energy. Despite temperatures theoretically sufficient for CO combustion and the potential for significant additional heat release (283 kJ/mol CO), plant measurements show that a significant amount of CO exits the shaft without being combusted, suggesting incomplete energy recovery, safety issues caused by explosions of trapped CO inside the suction line and high carbon emissions.



Figure 1.2: CO Explosion in a Steel Plant

The cause of incomplete CO combustion remains unclear. Possible reasons include insufficient oxygen from air infiltration, temperature decay along the shaft or poor mixing between reactants. Also, actual scrap and gas temperatures achieved during preheating are uncertain, since direct measurements inside the shaft are not practical.

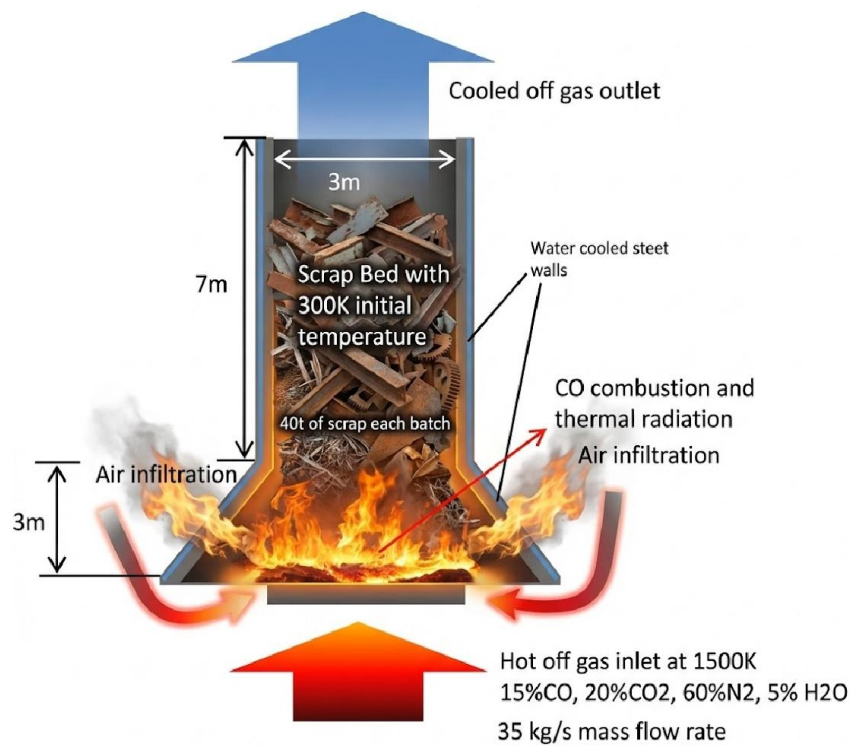


Figure 1.3: Scrap preheating operation inside the furnace shaft.

This thesis employs CFD modeling to investigate the coupled phenomena of heat transfer, fluid flow, and combustion in shaft preheating systems. The model simulates transient gas-solid heat exchange in the porous scrap bed, turbulent CO combustion with oxygen availability from air infiltration. By comparing scenarios with varying oxygen supply and operating conditions, the study aims to identify the mechanisms limiting CO conversion.

Research Objectives

The primary objective of this thesis is to develop a CFD model of EAF shaft preheating that identifies the fundamental mechanisms limiting carbon monoxide combustion and scrap heating effectiveness. Specific objectives include:

Model Development:

1. Calculate temperature-dependent thermophysical properties of EAF off gas mixtures (N₂, CO₂, CO, H₂O) using established correlations and mixing rules
2. Characterize the packed scrap bed through porosity, resistance coefficients (Ergun equation), and heat transfer correlations (Gunn correlation)

3. Implement thermal non-equilibrium modeling with separate gas and solid energy equations.
4. Validate mesh independence and convergence behavior

Combustion Investigation:

1. Model CO combustion using the Eddy Dissipation Model, validated through Damköhler number analysis
2. Quantify CO conversion rates and identify spatial distribution of combustion zones
3. Determine whether CO escape is caused by oxygen deficiency, thermal quenching, insufficient residence time, or poor mixing

Performance Assessment:

- Characterize transient temperature evolution in gas and solid phases during preheating cycles
- Determine actual final scrap temperatures and compare them with industrial assumptions
- Conduct parametric studies on off gas flow rate (5-45 kg/s) and particle size (0.05-0.30 m) effects on heat transfer.
- Quantify energy recovery efficiency including sensible heat transfer, combustion heat release, and heat losses

Process Understanding:

- Identify rate-limiting mechanisms preventing complete CO combustion
- Establish relationships between operating conditions (flow rate, pressure drop, air infiltration) and combustion performance

Methodology

This thesis employs computational fluid dynamics (CFD) to investigate heat transfer, fluid flow, and combustion in EAF shaft preheating systems using ANSYS.

The scrap bed is modeled as a porous medium with constant porosity of $\varepsilon = 0.85$ rather than modeling individual irregular scrap pieces. Flow resistance is calculated using the Ergun equation with viscous and inertial terms. Separate energy equations are solved for gas and solid phases (thermal non-equilibrium) coupled through convective heat transfer calculated via the Gunn correlation. The k- ε model captures turbulent mixing at high Reynolds numbers ($Re_p \approx 500-4300$).

CO combustion is modeled using the Eddy Dissipation Model (EDM), appropriate for mixing-limited reactions validated by Damköhler analysis ($Da \approx 5800$).

Simulation Methodology:

Transient simulations run for 450 seconds (7,5 minutes) starting when cold scrap (300 K) enters the shaft with hot gas (1500 K) flowing. This captures the batch preheating cycle where thermal equilibrium is not possible.

Boundary conditions:

- Inlet: Mass flow (5-45 kg/s), $T = 1500$ K, composition: 60% N_2 , 20% CO_2 , 15% CO , 5% H_2O (mass basis), with oxygen added for combustion cases
- Walls: Convective BC representing water-cooled panels ($T_{wall} = 450$ K)

Parametric Studies:

1. **Mass flow rate variation (5-45 kg/s):** Effects on heat transfer coefficients, gas exit temperature and scrap heating.
2. **Particle size variation (0.05-0.30 m):** Influence on specific surface area and heat transfer.
3. **Oxygen availability scenarios:**
 - No combustion (baseline sensible heat)
 - Stoichiometric O_2 ($1.2 \times$ theoretical)
 - Air infiltration due to pressure difference across the shaft

Thesis Organisation

The remainder of this thesis is organized as follows:

Chapter 2: State of Art and Science provides essential background on electric arc furnace (EAF), steelmaking technology and scrap preheating systems. This chapter reviews EAF operation, energy improvements through oxygen blowing and slag foaming, and existing scrap preheating methods including the Quantum shaft furnace investigated in this study. The chapter establishes the industrial context and identifies the knowledge gap regarding incomplete CO combustion in shaft systems.

Chapter 3: Methodology develops the computational framework for investigating scrap preheating and CO combustion. The chapter begins with mesh independence validation to ensure numerical accuracy, then systematically calculates temperature-dependent off-gas properties using VDI Heat Atlas correlations. Volume-averaged conservation equations for porous media are derived, followed by heat transfer characterization through the Gunn correlation and lumped capacitance validation via Biot number analysis. CO combustion modeling is established using the Eddy Dissipation Model, validated through Damköhler number analysis demonstrating mixing-limited reaction kinetics. Air infiltration calculations based on measured pressure drops complete the methodology, providing realistic oxygen supply estimates for combustion simulations.

Chapter 4: Results and Discussion present systematic investigation of scrap preheating and CO combustion optimization. The chapter progresses through four analytical stages: First, baseline scrap preheating characterization examines the effects of off-gas flow rate and particle size on heat transfer, revealing the discrepancy between predicted bulk temperatures and observed localized melting. Second, air infiltration studies demonstrate that stoichiometric oxygen becomes available at suction pressures as low as 800 Pa, yet CO still escapes, proving combustion is mixing-limited rather than oxygen-limited. Third, isolated air infiltration configurations reveal that front/back and side infiltration serve distinct complementary roles—front/back air creates center combustion zones while side air prevents wall CO escape.

Chapter 5: Summary synthesizes key findings and their implications for industrial design practice, emphasizing the shift from oxygen-limited to mixing-limited CO combustion in shaft preheating systems.

2. State of Art and Science

General Concept and Layout

- ① Scrap handling and logistics
- ② Scrap loading
- ③ Scrap charging
- ④ Scrap preheating
- ⑤ EAF steel melting
- ⑥ Power supply
- ⑦ Off gas treatment



Figure 2. 1 EAF Quantum Plant Layout

EAF Quantum Shaft Scrap Preheating System

The EAF Quantum shaft furnace is a vertical preheating system that uses hot off-gas from the melting process to preheat scrap metal before it enters the furnace vessel. The shaft is a fixed structure mounted on the furnace roof, designed to allow heat transfer from the rising off-gas to the scrap and recover off gas heat.

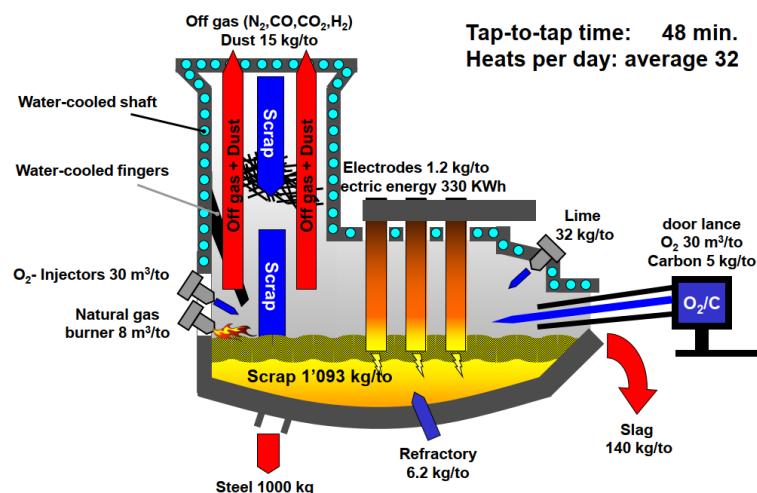


Figure 2.2 EAF Quantum Operation

Scrap metal is loaded into a chute at the loading terminal and elevated to the top of the shaft, where it is dumped onto closed fingers and preheated by hot exhaust gases. Once bucket melting is complete, the fingers open and the scrap drops into the furnace shell for melting. This charging sequence is repeated for two to four batches until the desired tap weight is reached, after which the steel is refined and tapped through a FAST-tapping system while the next scrap batch is already being prepared in the shaft.

Scrap Handling

The Scrap Handling System is built on a steel structure that supports a horizontal rail track. A shuttle frame with an attached scrap bucket moves along this track between the different loading points in the scrap yard and the Scrap Charging System of the Quantum EAF. The scrap buckets are filled by the scrap yard cranes.

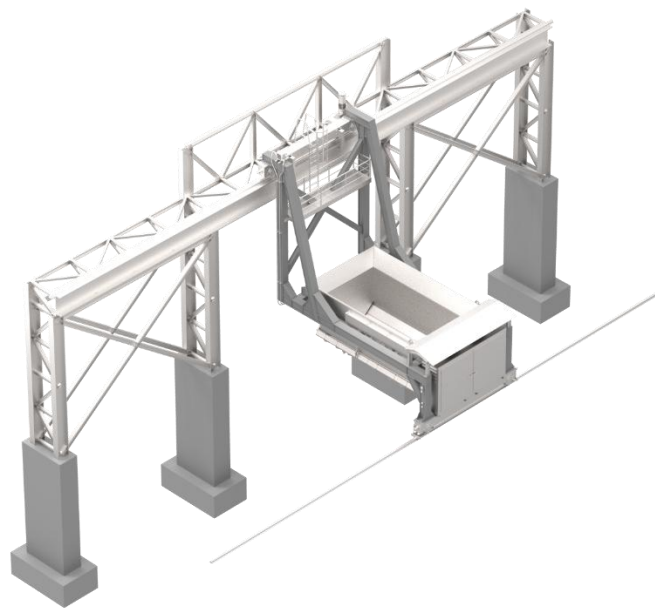


Figure 2.3 Scrap Loading System

Scrap Loading

The Scrap Charging System is installed on a steel structure that is fixed to foundations. The scrap container is placed on a shuttle that moves along inclined rails and is lifted by a winch system powered by an electric motor.

When the scrap is discharged from the shaft into the furnace the flaps open and the scrap container charges the next batch of scrap into the shaft.

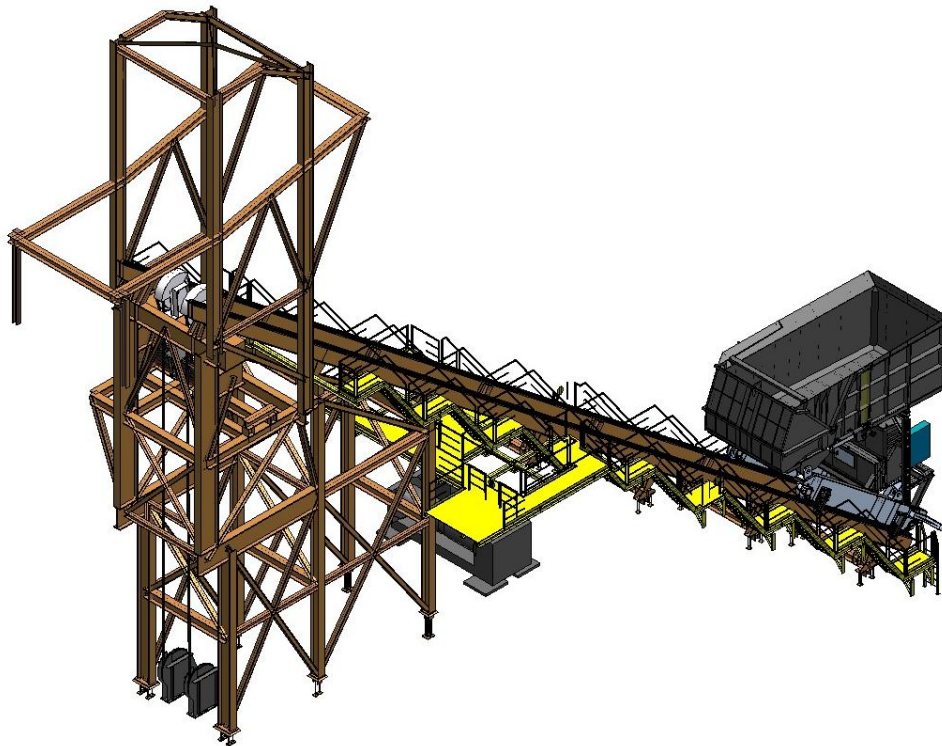


Figure2.4 Scrap Charging System

The container is tilted by a hydraulic cylinder built into the shuttle, allowing the scrap to be discharged into the shaft. After charging, the shaft flap closes. The container then returns to its original position on the shuttle, and the shuttle is lowered by the winch system for the next filling cycle.



Figure 2.5 Scrap Charging System Model

Scrap Charging

When the heat is tapped and the furnace is ready for the next cycle, the fingers are opened, and the scrap will be charged into the furnace from the shaft. Then the fingers are closed again, and the second batch of scrap is loaded into the shaft for preheating. Throughout the process the scrap is loaded into the shaft, preheated and charged into the furnace batch by batch.

Scrap Preheating

Scrap is the main metal charge for the electric arc furnaces, and this makes scrap preheating an important topic for energy efficiency improvement. Preheating scrap increases the enthalpy of the metal charge and reduces the total electricity and electrode consumption per ton of steel, therefore shortening melting times and improving productivity. Scrap preheating most effectively achieved using the thermal energy of EAF off gases, including both the sensible heat and the chemical energy of combustible components such as CO. [6]

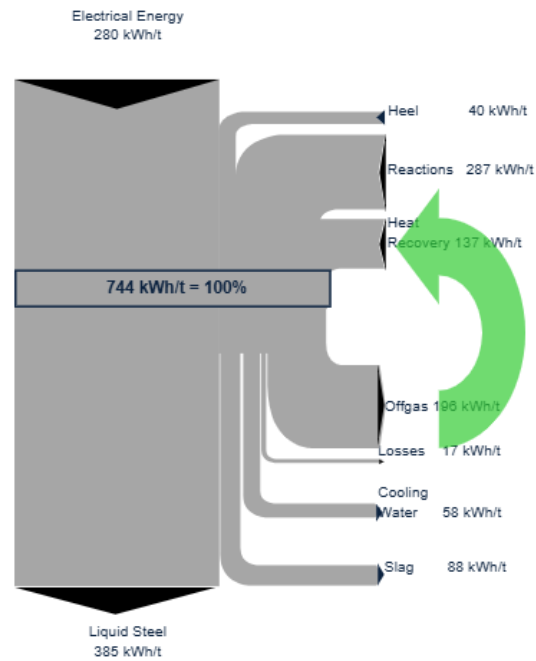


Figure 2.6 EAF Energy Balance Off Gas Heat Recovery

Shaft Geometry and Structure

The shaft has a trapezoidal cross-section that influences scrap distribution and off-gas flow patterns. The shaft structure consists of water-cooled panels to withstand thermal stress from the hot off-gas.

The top of the shaft is sealed by a water-cooled loading flap after the scrap batch is charged into the shaft. The loading flap guides the off gas to the exhaust elbows after it passed through the scrap and prevents the off gas from leaking to the environment. The flap is opened by hydraulic cylinders before the scrap charging and closed as soon as the scrap is charged.

Operating Sequence

1. **Charging:** Scrap is loaded from the top onto the closed finger system
2. **Preheating:** Hot off-gas flows upward through the packed scrap bed
3. **Heat Transfer:** Scrap is preheated for approximately 5-8 minutes; preheating is stopped if the charged scrap is molten. Then the next bucket must be charged to the steel bath

4. **Discharge:** Hydraulic cylinders retract the fingers, releasing the preheated scrap into the vessel

The system operates continuously with automated charging, eliminating the need for overhead cranes or roof opening during charging, which reduces heat losses and environmental emissions.

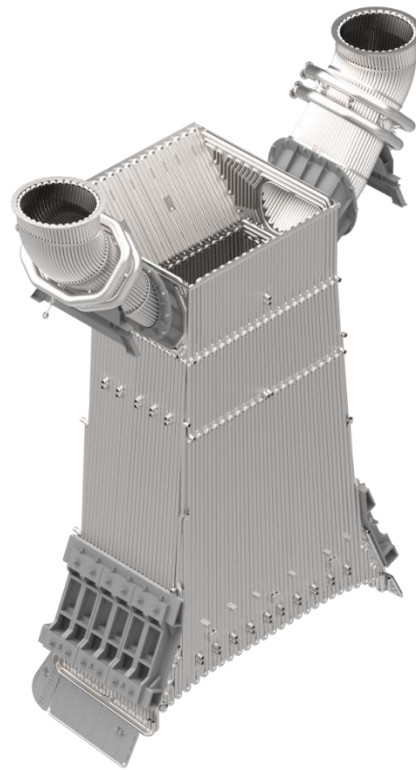


Figure 2.7 Shaft Design

The shaft connects to the primary suction system through two elbows located in the upper section, below the loading funnel. These exhaust openings on the side of the shaft guide the off-gas flow after it passes through the scrap bed.

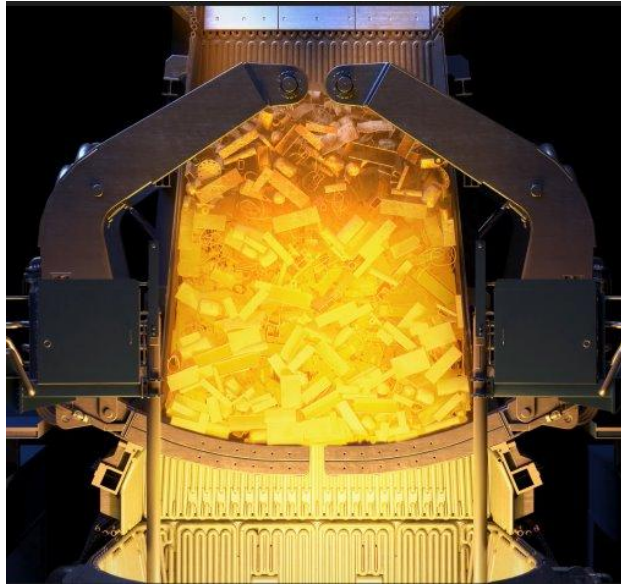


Figure2.2.8 Scrap Retaining System

Scrap Retaining System (Fingers)

At the bottom of the shaft, a retaining system consisting of water-cooled curved fingers holds the scrap during preheating. These fingers are mounted on steel beams moved by hydraulic cylinders, with each finger moving in an arc. During the charging of the shaft and the preheating process the fingers are closed. During the charging the scrap falls onto the fingers. There are spring absorbers between each finger and the holding structure that absorb the impact.

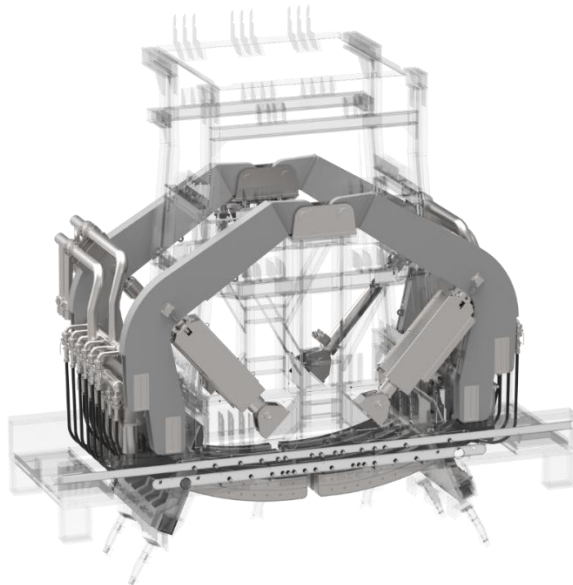


Figure 2.9: Finger Mechanism

When retracted, the fingers allow preheated scrap to fall into the furnace vessel (containing hot heel) in a manner similar to conventional bucket charging.

EAF Steel Melting

Electric arc furnace (EAF) uses electric energy for steel production by converting electrical energy into heat energy in an electric arc. EAF production accounts for nearly 30% of the 1.8 billion tons of steel produced annually. EAFs mainly use steel scrap that was wasted to produce useful steel [4].

Mechanical Design

Main parts of an EAF can be listed in 5 main parts:

- Lower shell: Eccentric Bottom Tapping (EBT), tap-hole, refractory lining.
- Hydraulic tilting platform
- Upper shell: Water cooled panels, furnace roof, off gas elbow, post combustion chamber.
- Power supply, gantry system with roof lifting and swing mechanism, electrode supporting arms, water cooled electrodes.
- Burner and injector systems.

The furnace is placed on concrete pedestals to raise the entire structure above the ground to enable a ladle to be driven underneath the furnace. On top of the pedestals there is the tilting platform which is a rectangular frame with hydraulic cylinders which makes it possible for the furnace to be tilted and the steel melt to flow out of the furnace through the taphole. The lower shell is a steel shell that is covered with refractory from the inside made of MgO-C bricks that protects the shell from the heat. The upper shell sits on top of the lower shell, and it is surrounded by water cooled panels from the inside [5].

Lower Shell

The lower shell of the EAF is a welded steel structure mounted on the tilting platform. It is lined with refractory material and designed to hold the full amount of liquid steel, including the tapping weight, hot heel, and the required slag volume.

The refractory lining is made of basic material with high MgO content and about 4% carbon. This composition helps protect the lining against slag damage and extends its service life.



Figure 2. 2.10 Lower Shell

Upper Shell

The furnace's upper shell has water-cooled panels, a slag door with tunnel, and a balcony panel. The panels are made from steel tubes, and water flows through them via the bezel ring.

A PLC monitors the cooling system, triggering an alarm or stopping operation if any critical value is exceeded. The monitored parameters shown on the display include:

- Inlet and outlet water temperatures
- Water flow rates
- Water pressure

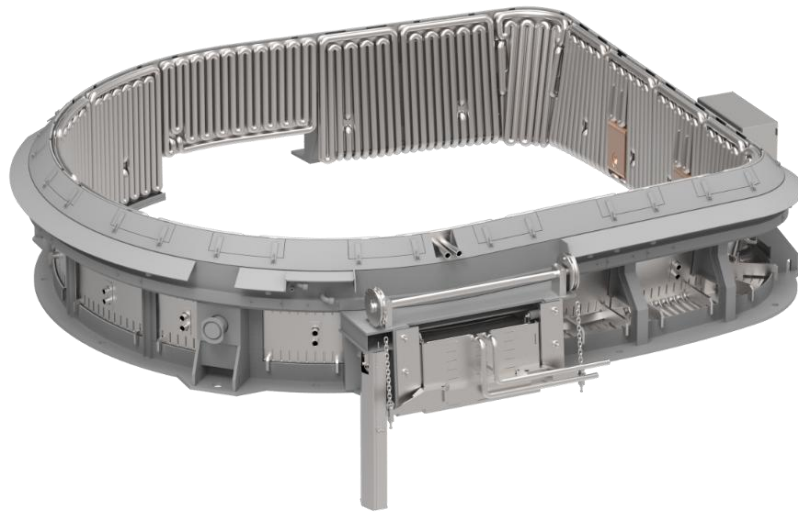


Figure 2. 2.11 Upper Shell

Furnace Roof

The furnace shell is sealed by a roof, which is fixed to a stationary support structure and connected to the shaft. The roof is water-cooled, with beams that hold the cooling elements. These beams also act as a collector ring (box or tubular design) for water supply and return. The roof supports the central roof piece and the connection adapter for the furnace shaft.

A furnace pressure measurement system on the roof enables closed-loop control of the off-gas damper. The sensor opening is automatically cleaned using a pneumatic punch controlled by the furnace PLC.

The roof also includes a water-cooled chute adapter for adding alloys or additives like carbon, lime, or DRI directly into the furnace near the arc.

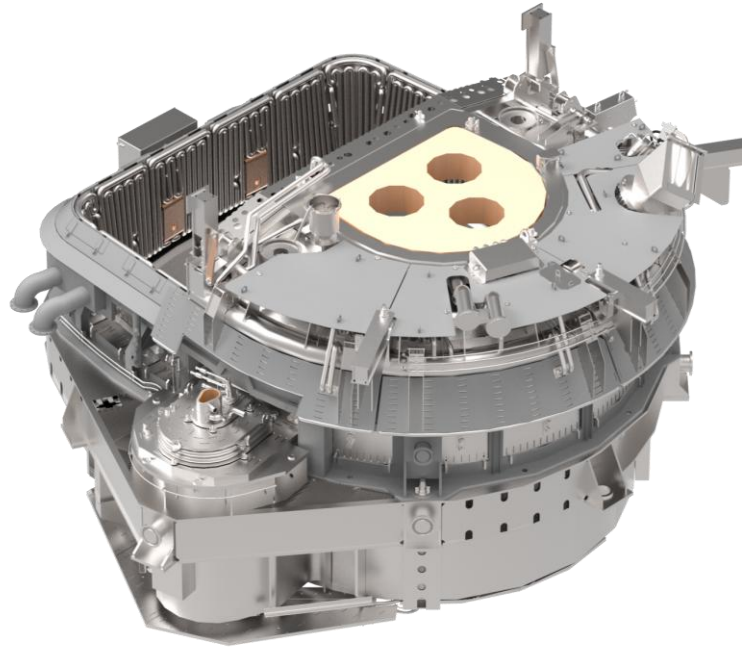


Figure 2.12 Furnace Roof

Tilting

The furnace tilts (shell lowering) using hydraulic cylinders. The tilt direction toward tapping or deslagging is controlled from the main control panel using a joystick. The tilting speed is set by the PLC. The furnace stays in its current position when the joystick is released.

The lower shell has a heat protection shield around the tap hole. Before starting the furnace, gunning is applied to protect the shell and shield from high heat during tapping. Temperature sensors monitor the lower shell continuously, positioned near the electrodes and tap hole.

Piping attached to the lower shell supplies compressed air and cooling water for the tap hole slide gate. Trunnions are provided on the lower shell so it can be lifted with the overhead melt shop crane.

Burner and Injector Systems

The center of the furnace has the highest melting power, but it might not reach near the furnace walls and cause cold spots. So additional fuel and oxygen are injected into the furnace near the walls to supplement chemical energy and help the melt heat evenly [5]. Oxygen also plays a role in metallurgical processes such as removal of phosphorus. The gas burners decrease electrical energy consumption and increase productivity by accelerating the melting process, but overall energy consumption doesn't change significantly. [3].

The increase in oxygen usage in steelmaking process has played an important role in improving the EAF productivity and reducing electrical energy consumption. This is due to the significant amount of chemical heat that is released when carbon, iron and other elements such as silicon and manganese oxidize, and this heat is absorbed by the molten bath.

Oxygen usage then increased even further with the injection of carbon powder. The oxygen that is blown into the bath causes an increase in the amount of oxidized iron and this causes a drop of yield. Carbon that is injected into the bath reduces iron oxides. In this process CO is produced which causes slag foaming.



Figure 2.13 Oxygen Top Lance

Oxygen and Carbon is injected into the steel bath with oxygen lances which are located right and left of the center piece. The tip of the oxygen nozzle allows the oxygen to be injected into the liquid steel with high momentum to penetrate the slag and transport the oxygen deeply into the liquid steel.

Slag Foaming

Slag plays a critical role in modern electric arc furnace operation by enhancing thermal efficiency, process stability, and metal yield. The foaming slag increases heat transfer efficiency from electric arcs to the molten bath by covering the arcs, reduces radiative heat losses and shields the walls and the roof of the furnace. Additionally reduces metal oxidation losses through FeO reduction and captures impurities such as phosphorus and sulphur.

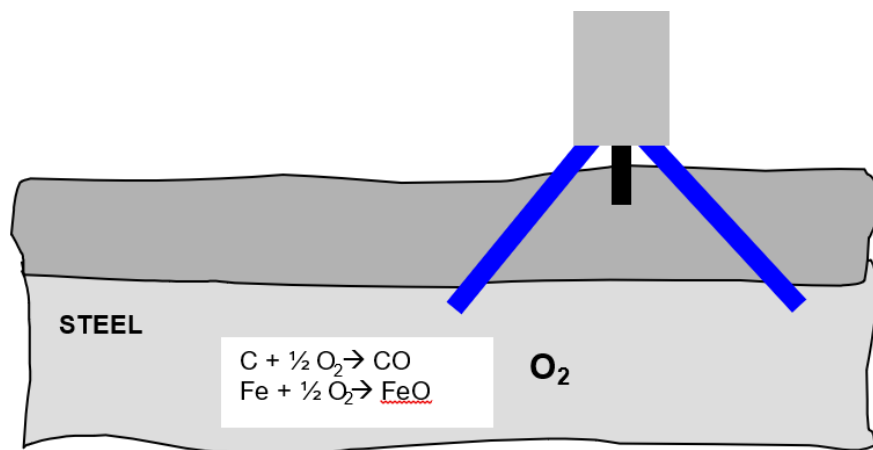


Figure 2.14 Slag Foaming

Oxygen oxidizes dissolved carbon in the bath, producing CO, while a portion of oxygen oxidizes iron to FeO. Injected carbon also reduces iron oxides ($\text{FeO} \rightarrow \text{Fe}$), generating additional CO. The combined effect of these reactions produces CO bubbles that foam the slag. Proper balancing of oxygen and carbon injection allows high oxygen usage without loss of yield. Through careful control of slag composition and foaming behavior, modern EAFs achieve improved productivity, reduce energy consumption, and more stable off gas and refining conditions.

Off Gas Treatment

The off gas is extracted from the system after it flows through the scrap pile inside the shaft into the direct fume extraction system, consisting of Post Combustion Chamber (PCC), Drop Out Box, Quenching tower and ducts to the bag house system.

The off gas from the shaft goes directly to the post combustion chamber where it is heated by gas burners. The design of the PCC allows sufficient residence time and uniform temperature distribution in the chamber for complete post combustion of CO and other pollutants.

After the PCC, the off gas passes through the drop out box where its velocity is reduced to allow large particles carried by the off gas to be separated from the off gas stream.

The top of the shaft is covered with a hood to collect the off gas that is escaping during the charging of the shaft through the open flap. Then the off gas that is collected by the hood on top of the shaft is extracted by the secondary dedusting system. Also the fumes that escape during charging and tapping as well as through the electrode ports are collected in the canopy hood that is above the furnace.

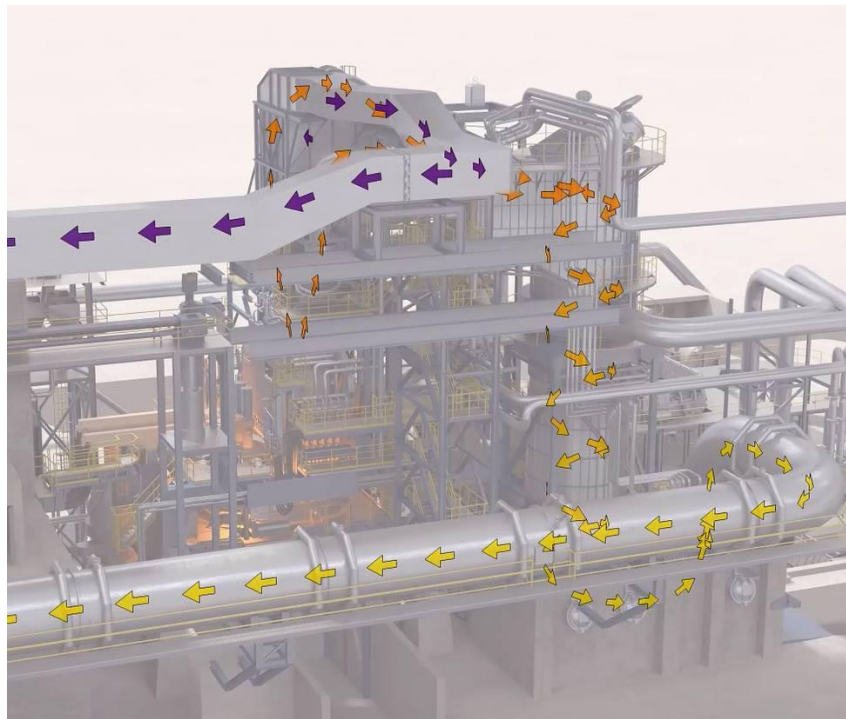


Figure 2.15 Off Gas Treatment

Scrap

Scrap is the main raw material for EAF steelmaking, providing metallic iron for the process. The EAF steel production route produces approximately 30% of the global crude steel. (Kirschen). Understanding scrap characteristics is essential for understanding the EAF operation, as scrap quality, composition and physical properties directly influence the process.

Steel scrap can be classified into two categories based on origin. First the postproduction scrap which is generated during steel manufacturing and processing operations such as casting, rolling, forging and machining. Second is the post-consumer scrap consisting of discarded products such as decommissioned industrial equipment, scrapped vehicles, obsolete machinery and tools and discarded household items.

Beyond origin classification, steel scrap is categorized by many physical characteristics. National and international standards define comprehensive classification systems based on form, dimensions, and bulk density. Bulk density is a critical parameter that affects melting behavior and furnace charging strategy, varying significantly depending on scrap form. High-density scrap (approximately 1.2-1.3 t/m³) includes thick steel plates (>10 mm thickness), heavy bars, and railroad rails. Medium-density scrap (0.8-1.0 t/m³) comprises tubular sections and structural pieces. Low-density scrap (0.7 t/m³) consists of thin sheet metal, wires, and mechanically compressed bales. The degree of corrosion is classified into three categories: degree I (not rusty with metallic surface), degree II (rusty surface with rust layer), and degree III (rusted-through or chemically damaged scrap) [4].

The chemical composition of post-consumer scrap causes challenges for high-quality steel production due to residual alloying elements that cannot be economically removed during EAF steelmaking. Typical residual element ranges in mixed scrap include copper (0.07-0.55%), nickel (0.03-0.20%), and chromium (0.04-0.18%) [EAF Fundamentals, Pretorius and Oltmann].], coming from electrical wiring, stainless steel components, and various alloyed steel grades. For production of high-quality specialty steels with strict residual element specifications, scrap is often supplemented with clean iron units such as direct reduced iron (DRI) containing less than 0.01% Cu, Ni, and Cr [3].

Before charging into the EAF, scrap goes through a preparation process that including segregation by type and density, shearing to appropriate dimensions, baling of low-density

materials for improved handling, and separation of nonferrous metals and nonmetallic contaminants. In shaft furnace applications, scrap preparation is particularly important as oversized pieces can cause flow blockages and uneven gas distribution through the packed bed [4].

In shaft furnace scrap preheating systems, the scrap forms a packed bed inside the shaft which allows the hot off gas to flow through in countercurrent arrangement. The physical characteristics of the scrap bed particularly porosity, particle size distribution, and bulk density—are critical parameters for heat transfer and fluid flow.

Porosity (ε), defined as the void fraction of the packed bed, is one of the most fundamental parameters which affects flow resistance and heat transfer in porous media. Experimental measurements of scrap bed porosity by Mandal (2010) for different scrap types showed porosity values ranging from 0.84 to 0.95 for shredded scrap varying with the distance from the walls of the volume in experiments. The measurements show that porosity approaches to 1.0 near container walls and reaches a uniform value in the bed core. For scrap with average particle length of 80 mm, the core porosity was measured at approximately 0.86-0.92. [19]

For the present study, a porosity of $\varepsilon = 0.85$ was selected as a conservative and volume averaged value to represent the packed bed in a furnace shaft that consists various scrap types.

The relationship between porosity and bulk density can be expressed as:

$$\varepsilon = 1 - (\rho_{\text{bulk}} / \rho_{\text{steel}})$$

where ρ_{bulk} is the scrap bulk density and $\rho_{\text{steel}} \approx 7850 \text{ kg/m}^3$ is the density of solid steel. For $\varepsilon = 0.85$, the corresponding bulk density is approximately 1180 kg/m^3 , which falls within the range of medium- to low-density scrap categories ($0.7\text{-}1.3 \text{ t/m}^3$) reported in scrap classification standards. [4]

Shaft Preheating Process and CO Post-Combustion Issues

1. Initial Furnace State (Before Scrap Discharge)

The furnace operates with under pressure, maintaining clean conditions with false air drawn in by suction. The system is stable, and post-combustion (PC) functions normally.

2. Scrap Discharge Phase

When the electrode is raised and the shaft flap opens, the under pressure is lost and off-gas begins escaping through the delta region. As scrap is discharged, a short but significant overpressure peak occurs flames escape through every gap in the furnace, and false air intake stops. When the fresh scrap drops into the furnace, an explosion occurs due to sudden overpressure.



Figure 2.16: CO Explosion due to Furnace Overpressure

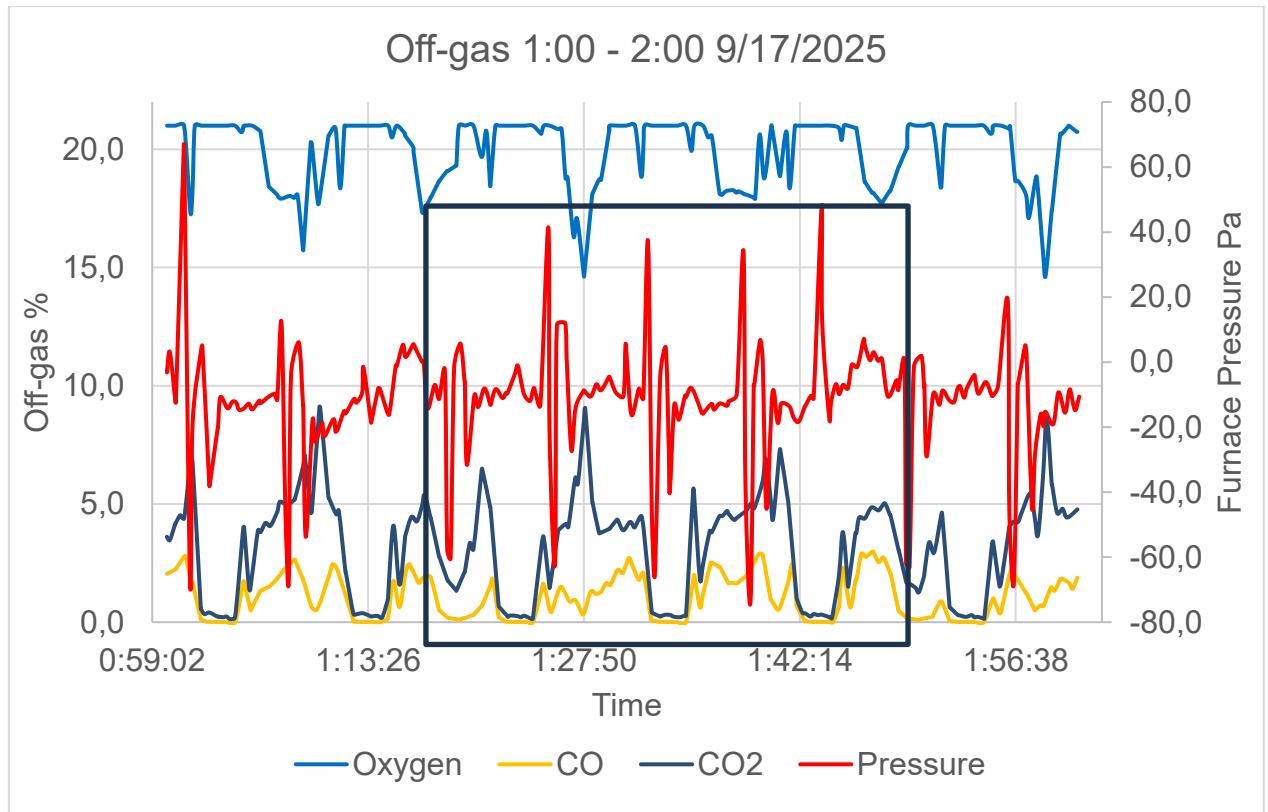


Figure 2.17: Plant Measurements of CO, CO₂, O₂ and pressure.

Figure 2.17 shows CO₂, CO, O₂ and pressure measurements. Peaks and drops in the pressure values show the under pressure and overpressure moments in the shaft. When the scrap is charged into the furnace it causes sudden overpressure pushing the off gas out of the furnace gaps which then combust as it contacts air outside the furnace. Then when the flap is opened for the next batch scrap charging into the shaft, open flap causes a chimney effect which causes under pressure. Due to the under pressure, off gas is sucked into the shaft with a large amount of false air and causes flames which can be seen at the shaft hood.

3. Empty Shaft Phase (Post-Discharge, Pre-Charging)

With the shaft now empty and the flap open, a strong chimney effect happens. The furnace returns to strong underpressure, and the shaft draws in large amounts of false air. Oxygen lances are reactivated, and carbon is injected to build a foaming slag. Due to the open shaft and chimney effect, suction is intense.

During this period, a pressure drop to around -15 mbar is observed. Offgas temperature increases, and CO combustion becomes visible in the hood.



Figure 2.18: CO Combustion at the Hood

4. Shaft Charging Phase

When scrap is charged into the shaft, it acts as a plug, cutting off the chimney-effect suction. This causes an overpressure buildup in the furnace. The off gas temperature rises, reaching a maximum of approximately 600°C . Once the flap is closed and damper is opened, temperatures drop down to 300°C and the suction of $380,000 \text{ Nm}^3/\text{h}$ is measured on the DEC at one of the steel plants. The off gas temperature then drops quickly to a stable $\sim 300^{\circ}\text{C}$, and the furnace pressure stabilizes at around -10 mbar.

5. Normal Melting Operation

During normal melting, the off gas temperature is around 300°C which is too cold for CO to ignite. The furnace process itself theoretically produces around $35,000 \text{ Nm}^3/\text{h}$ of off gas, but at the DEC it is measured to be $380,000 \text{ Nm}^3/\text{h}$ of actual flow. The difference is false air sucked in through gaps and leakages. This measured off gas flow rate is much more than the designed value. The main reason for this difference in designed value and measured value for off gas flow rate at this plant is that the air infiltration gap width is increased to 90mm from the default design value of 27mm.

The CO therefore passes through the shaft unburned and can find a source of ignition further downstream making this not just an efficiency problem, but also a safety concern.

7. Conclusions From Plant Observations

The core problem is a recurring cycle of overpressure events that suppress false air intake and therefore eliminate the oxygen supply needed for CO post-combustion:

- **Overpressure suppresses false air:** Any positive pressure event whether from scrap discharge, shaft charging, or rapid carbon addition reduces or eliminates false air drawn into the furnace. Without this oxygen, CO cannot combust.
 - **Reactions peak during the shaft discharge to closing the shaft:** Charging and discharging of scrap causes overpressure inside the furnace. Therefore the off gas and false air is not being sucked effectively, which then causes the off gas to flow out of the furnace from all the gaps and reactions occur out of the system.
 - **Slag reactions are not responsible:** No changes in behavior were observed before or after charging that could be linked to slag-steel reduction reactions.
 - **The reactions that are caused by organics:** Residues of organics on the dirty scrap are not being burned during the preheating and they increase the reactions at scrap discharge.
-

3. Methodology

Mesh Independence Study

Mesh Generation and Quality Assessment

Mesh independence study ensures that simulation results represent actual physics and there are no errors from the mesh discretization. The mesh divides the domain into small cells where equations are solved, and solution accuracy is dependent on the mesh size. The Ansys student license that was used in this study limits the maximum number of elements to one million thus the finest mesh consists slightly lower number of elements than one million. The medium mesh has the element amount of approximately half of the fine mesh and the coarse mesh have the half of the element amount of medium mesh. A mesh with insufficient amount of element size and low quality may give wrong predictions and an overly fine mesh wastes computational time. The mesh independence study finds the mesh size that is accurate and computationally efficient.

The domain was meshed using unstructured tetrahedral elements generated by ANSYS Meshing. Tetrahedral elements were selected for their ability to efficiently fill complex geometries.

In standard CFD simulations of external or internal flows, boundary layer mesh refinement near walls is critical for accurately solving velocity and temperature gradients in the viscous layer. However, the present study models the scrap bed as a porous medium rather than resolving individual scrap particles, which changes the meshing requirements.

In porous media modeling, the momentum source terms (Ergun equation) effectively represent the drag forces caused by the solid particles on the fluid flow through the entire volume. The flow resistance is distributed uniformly according to the specified porosity and resistance coefficients, rather than being concentrated at discrete solid surfaces. Thus, there are no thin viscous boundary layers adjacent to individual particles that require fine mesh resolution.

Similarly, heat transfer between gas and solid phases is modeled through a volumetric source term distributed throughout the porous zone, rather than through boundary conditions at particle surfaces. The heat transfer coefficient is a volume averaged parameter that

accounts for the total heat exchange between phases without requiring resolution of thermal boundary layers around individual particles.

k - ϵ turbulence model with standard wall functions is employed in this study. For porous media applications, the turbulence model parameters (k , ϵ) are volume averaged. The model does not solve turbulence near individual particle surfaces, but it captures the bulk turbulence intensity and eddy dissipation within the void spaces.

In standard fluid domains, wall functions require specific y^+ values which is a dimensionless number that describes how far the center of the first mesh cell is from a wall surface however in porous media the porous zone is modeled as a bulk without discrete wall surface thus the y^+ concept is not relevant. Also, the momentum and energy exchange happens through distributed source terms rather than wall boundary conditions. For these reasons wall boundary refinement was not employed.

The shaft walls with water cooled panels have boundary conditions but these surfaces are not the main interest for mesh sensitivity analysis. The temperature gradients next to these walls are relatively small compared to vertical temperature gradients within the packed bed and the uniform mesh produces decent results for the wall heat transfer.

The mesh quality characteristics are summarized in Table 3.1. All three meshes satisfy Ansys recommended quality standards (orthogonal quality > 0.15 , skewness < 0.85). The orthogonal quality improved slightly with mesh refinement from 0.79 (coarse) to 0.83 (very fine), while the average skewness decreased from 0.20 to 0.16, indicating better element alignment with the expected flow direction. The mesh refinement strategy progressively reduced the characteristic element size from 0.154 m in the coarse mesh to 0.069 m in the finest mesh.

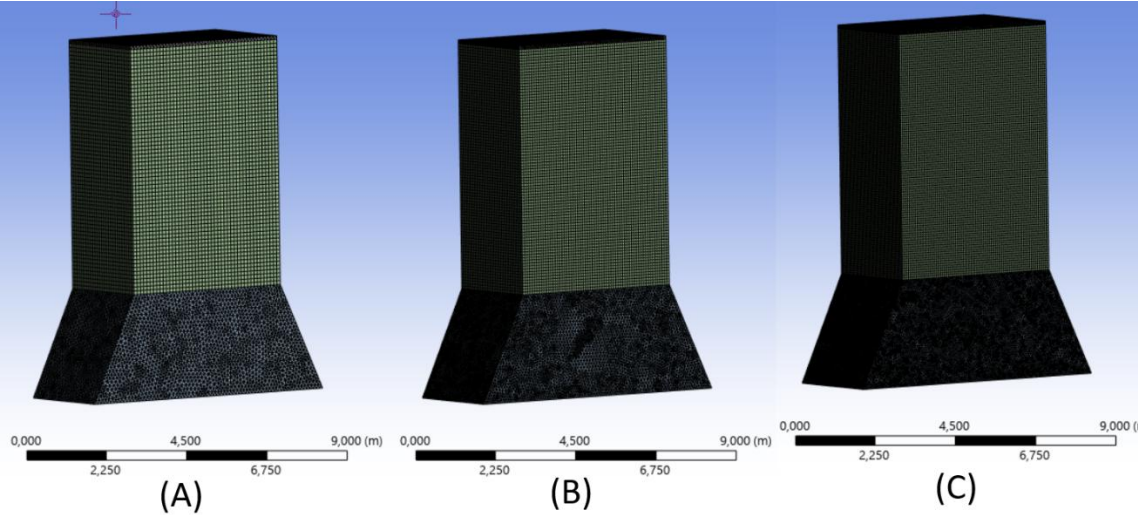


Figure 3.1: Mesh Refinement

The figure 3.1 shows the low mesh (A), medium mesh (B) and the fine mesh (C).

		Coarse	Medium	High
	Elements	250504	499887	996551
	Nodes	71631	175527	390991
	Size	0,154	0,096	0,0718
Orthogonal	Minimum	0,25	0,25	0,25
	Maximum	1	1	1
	Average	0,79	0,81	0,83
Standard Deviation		0,13	0,13	0,13
Skewness	Minimum	1,30E-10	1,30E-10	1,30E-10
	Maximum	0,75	0,75	0,75
	Average	0,2	0,18	0,16
Standard Deviation		0,134	0,138	0,14

Table 3.1: Mesh Quality Metrics

Table 3.1 Shows the element sizes and total number of elements with corresponding quality metrics for each mesh.

Solution for Mesh Independence Study

Each mesh configuration was simulated using identical physics models, boundary conditions, and solver settings:

Solver Configuration:

- Solver type: Pressure-based, transient
- Turbulence model: Realizable k- ϵ with standard wall functions
- Porous media: Viscous resistance $1/\alpha = 550 \text{ m}^{-2}$, inertial resistance $C_2 = 8.55 \text{ m}^{-1}$
- Thermal model: non-equilibrium, dual cell (separate gas and solid temperatures)
- Time step: $\Delta t = 2 \text{ s}$
- Total simulation time: 300 s (5 minutes of preheating)
- Convergence criteria: Scaled residuals $< 10^{-4}$ for all equations

Boundary Conditions:

- Inlet: Mass flow inlet, $\dot{m} = 45 \text{ kg/s}$, $T_{\text{gas}} = 1500 \text{ K}$
- Outlet: Pressure outlet, $P = 101,325 \text{ Pa}$ (atmospheric)
- Initial conditions: $T_{\text{gas}} = 1500 \text{ K}$, $T_{\text{solid}} = 300 \text{ K}$

Monitored Parameters

Five key parameters were monitored to assess grid independence, representing critical physical quantities for the scrap preheating process:

1. **Minimum gas temperature** (Gas Min T): Minimum temperature in the gas phase [K]
 2. **Average gas temperature** (Ave gas T): Volume-averaged gas temperature [K]
 3. **Maximum scrap temperature** (Scrap Max T): Maximum temperature in the solid phase [K]
 4. **Minimum scrap temperature** (Scrap Min T): Minimum temperature in the solid phase [K]
 5. **Average scrap temperature** (Ave scrap T): Volume-averaged solid temperature at $t = 300 \text{ s}$ [K]
-

These parameters capture both volume-averaged integrated quantities and local extremes, providing assessment of mesh sensitivity.

Results and Grid Convergence Analysis

The results that are obtained from each mesh are summarized in Table 3.2 demonstrating reasonable behavior with progressive mesh refinement. (45kg/s off gas flow rate, $d_p = 0,1\text{m}$, $t=300\text{s}$)

45kgs 0,1m	Gas Min T (K)	Ave gas T (K)	Scrap Max T (K)	Scrap Min T (K)	Ave scrap T (K)
Coarse	680	1026	849	433	598
Medium	692	1036	834	433	596
High	700	1043	819	435	592

Table 3.2: Mesh Independence Results

To quantify the grid convergence, relative changes between successive mesh refinements were calculated:

From Coarse to Medium mesh:

- Gas minimum temperature: +1.8%
- Gas average temperature: +1.0%
- Scrap maximum temperature: -1.8%
- Scrap minimum temperature: 0.0%
- Scrap average temperature: -0.3%

From Medium to Very Fine mesh:

- Gas minimum temperature: +1.2%
- Gas average temperature: +0.7%
- Scrap maximum temperature: -1.8%
- Scrap minimum temperature: +0.5%
- Scrap average temperature: -0.7%

The results indicate that:

1. Gas temperatures show increasing trend with mesh refinement, with changes decreasing from the coarse to very fine mesh. The gas minimum temperature demonstrates the largest sensitivity, increasing by 20 K (2.9%) from coarse to very fine mesh, while the gas average temperature increases by 17 K (1.7%).
2. Scrap maximum temperature shows a decreasing trend with mesh refinement, decreasing by 30 K (3.5%) from coarse to very fine mesh. This suggests that coarser meshes slightly overpredict peak scrap temperatures.
3. Scrap average temperature shows excellent convergence, with changes of less than 1% between successive refinements and a total variation of only 6 K (1.0%) across all meshes. This is the primary parameter of interest for assessing preheating effectiveness.
4. Scrap minimum temperature remains constant across all meshes (433-435 K), indicating that the inlet region behavior is well-captured even with the coarse mesh.

Mesh Selection

Based on the mesh independence study, the medium mesh with approximately 500,000 elements was selected to balance accuracy and computational efficiency. This selection was justified as follows.

1. The average scrap temperature shows less than 1% variation between medium and fine mesh.
 2. The medium mesh captures essential physics of scrap preheating process with acceptable accuracy in much shorter time. The medium mesh required approximately 250 minutes where the fine mesh takes 830 minutes.
 3. All monitored parameters show that the solution is mesh independent, with changes between the medium and fine mesh below 2%.
-

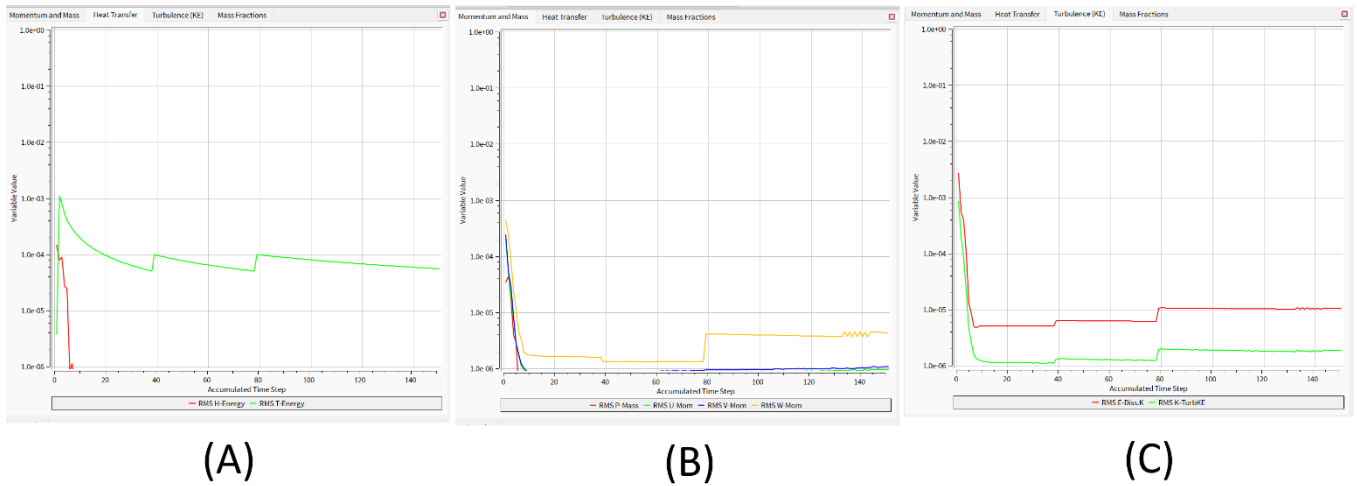


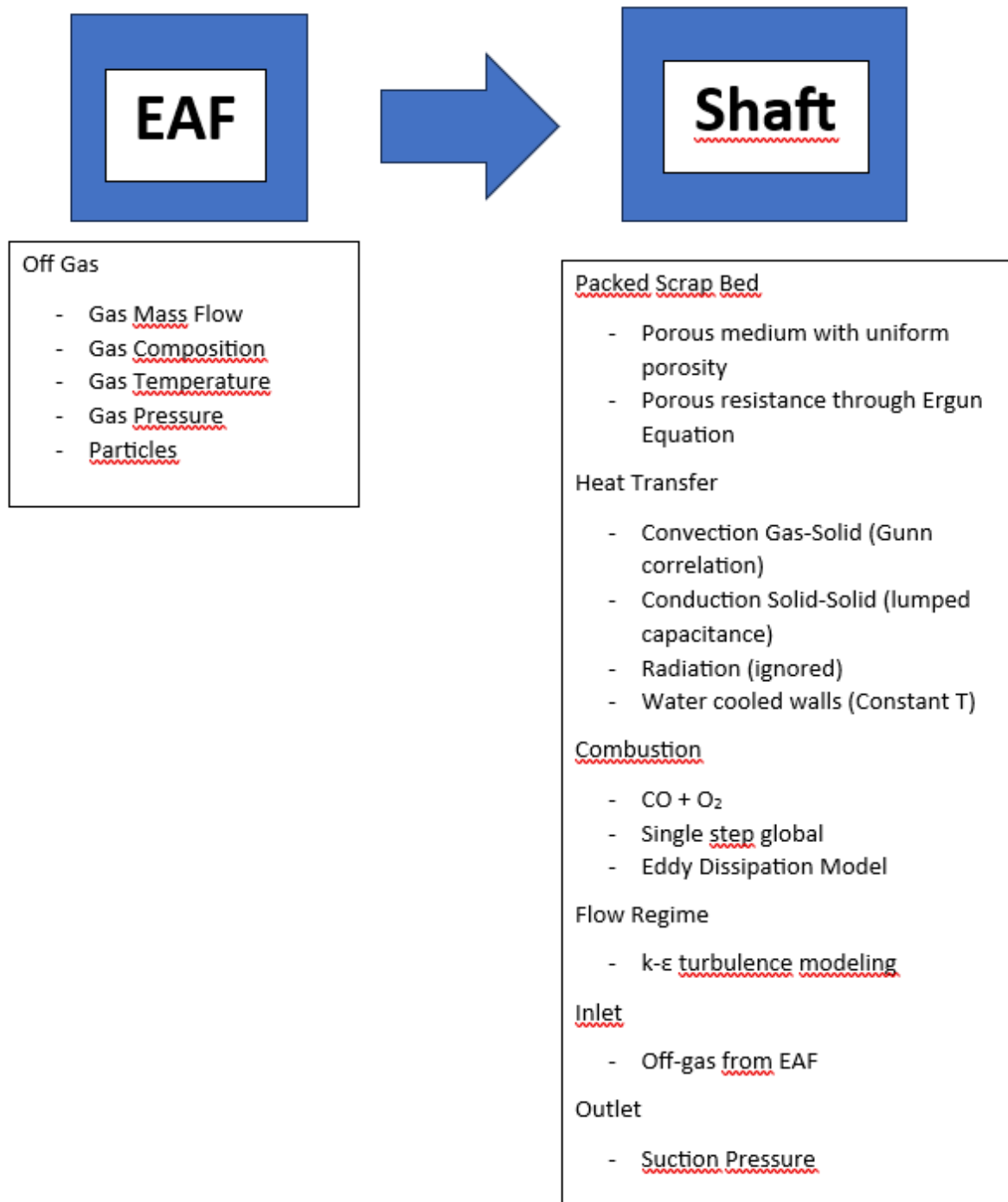
Figure 3.2 : Residuals Plot

Figure 3.2 presents representative residual convergence histories across the 300-second transient simulation for the coarse mesh case. The plots show that all governing equations—continuity, momentum (x, y, z components), energy (gas and solid phases), and turbulence (k- ϵ model) achieve convergence within 15-20 iterations per time step.

The energy equation residuals, which are critical for thermal predictions, converge to approximately 10^{-5} for the gas phase and 10^{-4} for the solid phase. Momentum and continuity equations achieve even stronger convergence (10^{-6}), while turbulence model equations stabilize at 10^{-5} . All values are well below the specified convergence criterion of 10^{-4} for scaled residuals.

Mesh independence study was performed following established CFD verification method (Eça & Hoekstra). The solution shows stable, near monotonic convergence. Minor increases or oscillations in residual values (visible around time steps 40 and 80) correspond to data file output operations and immediately return to converged levels without affecting solution accuracy. No convergence difficulties or numerical instabilities were encountered throughout the simulation, confirming adequate numerical stability for the selected discretization and solver settings. [12]

Boundary Conditions



Overview and Parameter Identification

Establishing physically representative boundary conditions is critical in CFD simulation of EAF shaft preheating systems. As Odenthal et al. note, reliable boundary conditions for EAF off-gas systems cannot be specified a priori; they necessarily depend on empirical plant data and measured values, since gas mass flow rate, temperature, chemical composition, pressure, and particulate loading all vary substantially over the course of an operating cycle and between different process phases. The present work identifies the following parameters as governing the inlet gas condition to the packed scrap bed:

- Volumetric/mass flow rate of the off-gas entering the shaft
- Static pressure (relative to atmosphere) at the shaft inlet, which drives gas infiltration
- Inlet gas temperature
- Chemical composition (CO, CO₂, O₂, N₂, H₂O mass fractions)
- Particulate (dust) loading

These parameters are interconnected: the pressure difference between the EAF furnace interior and the dedusting system drives the off-gas flow rate, which in turn determines residence time, combustion extent, and heat transfer to the scrap. The composition is tightly coupled to the EAF process phase, since CO generation rates and oxygen injection schedules differ fundamentally between the flat-bath melting phase and the refining phase.

Operating Modes

- Empty shaft – Charging
 - Preheating
 - Discharging
 - Melting
 - Tapping
-

Off Gas

The variability in reported off-gas mass flow rates and compositions is caused by several factors that differ between plants and even between heats at the same plant.

False air infiltration through the slag door, electrode ports, and furnace gaps plays a big role. Since the furnace operates under slight negative pressure, the amount of infiltrated air depends on suction power and flap conditions, directly diluting CO and raising O₂ in the measured off-gas. Post-combustion in the furnace also effects the composition by available oxygen, mixing quality, and slag foaming level.

Operational differences also play a strong role. Oxygen lance timing, carbon injection for slag foaming, and decarburization intensity all vary between plants and process stages, causing CO concentrations to change dramatically within a single heat. Finally, furnace size influences gas residence time in the furnace and the scale of all the other effects.

	N ₂ %	CO ₂ %	CO %	O ₂ %	H ₂ O %	H ₂ %	Off Gas T	Flow Rate
SMS group GmbH, Internal Information, 2017.	58,5	20	18	3,45		0,05	2173K	56000Nm ³ /h
Arzpeyma et al.	-	15	25	17,7	-	-	1800K	
Jacqueline Christina Gruber	14,7	70,3	5,9	2,1	7	-	1873K	13,64kg/s
T. Meier, K. Gandt, T. Echterhof, H. Pfeifer	-	25-30	20-50	-	-	-	1500-2000K	
Karbowniczek and Kawalkowski	-	19.3	19	-	-	-	1893K	
Kirschen		10-25	50-70	-	-	-	~1473 K	~100000 Nm/h
Primetals Internal	60	20	15	0	5	0	1500K	100000 Nm/h

Table 3.3: Off Gas Parameters

Table 3.3 summarizes reported EAF off-gas compositions, temperatures and flow rates from the literature. Values attributed to Karbowniczek and Kawalkowski, Arzpeyma et al., and SMS Group GmbH were not obtained from their original publications but are taken from the review of Odenthal et al. (2018), where they are cited as boundary conditions

used in CFD post-combustion studies. The remaining entries are drawn directly from the cited publications.

Another reason for the variation of off gas species concentration and flow rates is that the measurements or calculations are not performed consistently at a single spot. Off gas measurements taken after the elbow show lower CO but higher CO₂, N₂ and O₂ due to the false air from the elbow gap. For this study it is considered that the false air leaks into the furnace from various gaps therefore the off gas also consists of N₂, to account for this variations sensitivity study of CO concentration is performed.

Gas Inlet Temperature

The off-gas temperature at the point of entry into the scrap bulk is one of the most influential boundary conditions. In the literature off-gas temperatures of 1600-1900K have been reported for conventional EAFs. These values usually represent the off-gas temperatures after post-combustion at the elbow due to air infiltration. In shaft furnaces post combustion of CO rich off gas takes place inside the shaft under the scrap region due to air infiltration from the gaps on the shaft walls. For the current CFD model, gas temperature for inlet boundary condition must represent the temperature of the off gas coming from the furnace before post combustion of CO, which means a lower temperature needs to be selected. By the time gas reaches the base of the shaft scrap bed, some cooling has occurred through radiation losses and partial mixing with secondary air. Based on plant measurements reported by Kirschen et al. for EAF off-gas at the elbow, temperatures up to 1650 °C occur during oxygen injection peaks, with typical values during steady melting in the range of 800–1200 °C. Therefore, an off-gas temperature of 1500K is selected for this study.

Shaft Outlet Boundary (Suction Pressure)

Flow through the shaft scrap bed is driven by the negative pressure (suction) maintained by the dedusting fan system. This pressure drop across the shaft is the primary control variable for off-gas flow. From the plant data suction pressure of 1000-1200Pa is taken for the base-line case where the flap is closed and scrap is charged into the shaft. When flap is opened for charging the scrap into the shaft, the seal is broken therefore the pressure at the shaft outlet is equal to the ambient pressure. In the plant flames are observed outside the shaft

inlet area and to understand the mechanism that causes it, simulations with outlet pressure equal to ambient pressure is made.

Gas Mass Flow Rate

Gas mass flow rate through the shaft varies in measurements from plant to plant and in literature. The reason for that is the air infiltration into the shaft varies from plant to plant due to differences in the width of air infiltration gaps, operational differences and potentially any holes or gaps on the suction line specific to the plant. A volumetric flow rate of 100000 Nm³/h which corresponds to 35kg/s is chosen as the baseline based on Primetals internal data. To account for the effects of variation in the mass flow a parametric study for varying mass flow rate from 5kg/s to 45kg/s is conducted.

Chemical Composition

The chemical composition of EAF off-gas is highly variable across the operating cycle. During active decarburization, CO concentrations at the EAF elbow reach 50–70 vol.-% (Kirschen et al.). After partial combustion in the off-gas duct and secondary air dilution, compositions entering the shaft region are significantly moderated. Two representative operating points reported for a 140 t AC EAF by Karbowniczek and Kawalkowski illustrate the range: an oxygen-lean state (19% CO, 19.3% CO₂, 0% O₂) and an oxygen-rich state (3.2% CO, 1.1% CO₂, 17.7% O₂). For a 180 t ConArc EAF, Meier et al. report primary gas compositions of 18% CO, 20% CO₂, 3.45% O₂, 58.5% N₂ at 1900 °C as representative inlet conditions. In this study, 15% CO mass fraction is chosen as baseline but to account for the effect of varying CO concentrations a parametric study from 15% to 50% of CO concentration is conducted.

Scrap Bed Initial Temperature

The scrap is assumed to enter the shaft at an initial temperature of 300K (27°C), corresponding to the ambient conditions. Prior to charging into the shaft, the scrap is stored outside therefore, in thermal equilibrium with the surrounding environment. No preheating of the scrap occurs before it enters the shaft. Any seasonal variation in ambient temperature is considered negligible for this study.

To account for these variations in boundary conditions, 5 simulation groups from A to F are defined. Each group consists of several simulation cases, where boundary condition parameters are changing according to their group. Since the exact values are not possible to obtain, varying these parameters makes it possible to quantify the sensitivity of the system to these parameters. These changing parameters are:

1. Mass flow rate
2. Suction pressure
3. Off gas CO concentration
4. Scrap
5. Flap position
6. Air infiltration gap width

Group A represents the operating conditions where scrap is being charged into the shaft. It is divided into 3 sub-groups that consists of 3 cases. First sub-group from A1 to A3, represents the operation step where the flap is opened before the scrap is charged. The aim of this 3 cases is to understand the off gas behavior when the flap is open. In plant observations it is seen that the off gas leaks out of the shaft and combusting as soon as it contacts the air. Three cases were conducted with different mass flow rates to understand the relationship between the variation of mass flow rate due to the varying species production rate inside the furnace and the amount of escaping off gas to the environment.

A	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
A1	35	15	open	0	no	27
A2	40	15	open	0	no	27
A3	45	15	open	0	no	27
A4	35	15	closed	-1000	no	27
A5	40	15	closed	-1000	no	27
A6	45	15	closed	-1000	no	27
A7	35	15	closed	-1000	yes	27
A8	40	15	closed	-1000	yes	27
A9	45	15	closed	-1000	yes	27

The second sub-group from A4 to A6 represents the phase after the scrap is discharged into the furnace before the flap is opened for the next batch of scrap charging into the shaft. In the real plant the flap is opened before the charging and it stays open until the scrap is fully charged into the shaft but to capture the chimney effect that occurs in plant operation in the simulations, negative pressure from the outlet is defined. The negative pressure value is defined as -1000Pa for the cases that represent this charging phase (A4, A5, A6) and normal operation phase (A7, A8, A9) therefore it is not expected to find the actual difference in false air flow rates of these two cases. However it is important to simulate the scrap charging phase with this given simplification to understand the improved CO combustion during the charging due to the chimney effect.

The third sub-group from A7 to A9 represents the phase after the scrap charged into the shaft for preheating. This two sub-groups are compared to understand the off-gas behavior inside the shaft with and without scrap. This comparison shows how the scrap affects the gas velocities and mixing of off-gas with the infiltrating air.

B	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
B1	35	15	closed	-600	yes	27
B2	35	15	closed	-800	yes	27
B3	35	15	closed	-1000	yes	27
B4	35	15	closed	-1200	yes	27
B5	35	15	closed	-1400	yes	27
B6	35	15	closed	-1600	yes	27
B7	35	15	closed	-1800	yes	27
B8	35	15	closed	-2000	yes	27

Group B consists of 8 cases at constant mass flow rate, CO concentration, gap width with scrap present and flap closed. In this group suction pressure is varied to quantify the infiltrating air at different suction pressures. The aim of this group is to compare the oxygen that is introduced into the system through the air infiltration gaps due to suction at the outlet, with the calculated stoichiometric oxygen requirements for the off-gas with a mass flow rate of 35kg/s and 15% CO concentration to understand if over stoichiometric oxygen can be supplied by increasing the suction pressure and if complete CO combustion can be achieved due to increased oxygen content and better mixing due to increased air flow.

Group C consists of 8 cases at constant mass flow rate, gap width with scrap present and flap closed. The CO concentration is increased at each case starting from 15% to 50% with 5% increments. The CO concentration can vary due to the differences in operation of oxygen and carbon lances, therefore relying on a single representative CO concentration would risk drawing conclusions that are only valid for a narrow subset of real operating conditions. By spanning the full practically relevant range, Group C allows the CO combustion rate to be evaluated as a function of inlet concentration under identical flow and mixing conditions, thereby isolating the effect of fuel availability from other influencing parameters. If simulations at low CO concentrations show near-complete combustion, this confirms that the available oxygen is sufficient under those conditions; conversely, cases at high CO concentrations reveal whether oxygen availability or mixing becomes the limiting factor as the fuel load increases.

C	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
C1	35	15	closed	-1000	yes	27
C2	35	20	closed	-1000	yes	27
C3	35	25	closed	-1000	yes	27
C4	35	30	closed	-1000	yes	27
C5	35	35	closed	-1000	yes	27
C6	35	40	closed	-1000	yes	27
C7	35	45	closed	-1000	yes	27
C8	35	50	closed	-1000	yes	27

Group D consists of 4 cases at constant mass flow rate, with scrap present and flap closed. The default air infiltration gap in the 150t Quantum furnaces is 27mm but in reality some plants increase this gap size to allow higher air flow into the shaft to improve post combustion of CO. Group D aims to quantify the relationship between the gap width and infiltrating air.

D	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
D1	35	15	closed	-1000	yes	27
D2	35	15	closed	-1000	yes	54
D3	35	15	closed	-1000	yes	81
D4	35	15	closed	-1000	yes	108

Group E consists of 4 cases where mass flow rate, gap width and CO concentrations are kept constant. Cases are simulated with different suction pressures from 1000Pa to 2000Pa with 200Pa increment at each case. Group F also aims to investigate the effects of increasing air flow due to increase in suction pressure on combustion and scrap preheating with CO rich off-gas.

E	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
E1	35	15	closed	-500	yes	54
E2	35	15	closed	-1000	yes	54
E3	35	15	closed	-1200	yes	54
E4	35	15	closed	-2000	yes	54

Group F consists of 3 cases with constant mass flow rate, CO concentration and suction pressure. Gap width is kept at 27mm as default but air infiltration is not allowed from all gaps. By allowing air infiltration from first the sides and then the back and front section this group of cases aims to investigate how air infiltration from different sides effect the mixing and combustion inside the shaft.

F	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
F1	35	15	closed	-1000	yes	27-all sides
F2	35	15	closed	-1000	yes	27- front/back
F3	35	15	closed	-1000	yes	27-only sides

Off gas Properties

The off gas that come out of the furnace and flowing through the scrap in preheating is a mixture of mainly CO, CO₂, N₂, H₂O. To accurately predict the heat transfer and pressure drop temperature dependent properties of the gas composition such as density, viscosity, thermal conductivity and specific heat capacity, needs to be evaluated from individual component properties.

Gas	Molar Mass (kg/mole)	Mass Fraction %	Mole Fraction
N ₂	0,028	60	62,81
CO ₂	0,044	20	13,32
CO	0,028	15	15,70
H ₂ O	0,018	5	8,14

Table 3.4: Gas Compositions

Table 3.4 demonstrates the molar mass of individual gases, the individual mass and molar fractions of each individual species that makes the off gas.

Gas Composition

The composition of the gas and the ratios of the gases vary from operation to operation, depending on the scrap, air infiltration, and additives into the melt. In literature slightly different ratios and compositions can be found. In this section representative values of off gas composition is adopted that are in alignment with literature and plant data, 15%CO, 20%CO₂, 5%H₂O, 60%N₂ and the methods to calculate the mixture properties are presented. It is assumed that the off gas doesn't contain any oxygen but for the CO combustion simulations it is going to be modeled as air infiltration from gaps based on pressure drop and oxygen addition based on stoichiometric calculation where the required amount of oxygen is calculated based on the CO content of the off gas. [5]

Molecular Weight of Gas Mixture

Molecular weight of the gas mixture is a fundamental value for calculating gas density through the ideal gas law, which directly affects all momentum transport calculations, and it can be calculated as:

$$M_{mix} = \sum_{i=1}^N x_i M_i \left(\frac{kg}{kmol} \right) \quad 3.1$$

where x_i is the mole fraction and M_i is the molecular weight of each gas.

Density of Gas Mixture

The superficial velocity of the gas and Reynolds number is dependent on the gas density therefore it is an essential property to calculate. Density is also a property that is temperature dependent, and it can be calculated using the ideal gas law:

$$\rho_{mix} = \frac{PM_{mix}}{RT} (kg/m^3) \quad 3.2$$

Where P is the absolute pressure (101,325Pa), R is the universal gas constant= 8.3144 [kJ/(kmol.K)] and T is the temperature (1500K).

Dynamic Viscosity of Gas Mixture

Dynamic viscosity is also another property that affects the Reynolds number, which in turn affects the Nusselt number and heat transfer coefficient through the Gunn correlation. It also affects the viscous component of Ergun equation. That makes the accuracy of the dynamic viscosity calculation very critical.

To calculate the dynamic viscosity of the gas mixture a semi empirical method is used called Wilke's mixing rule [13]. This method accounts for the interactions of different molecules from different species. Different molecules with different mass and mobility collide and exchange momentum with each other differently so rather than simply averaging viscosities Wilke's method captures these interactions by weighting factors that depend on the viscosity and molar mass ratios of components.

Wilke's mixing rule:

$$\mu_{mix} = \sum_i \frac{x_i \mu_i}{(\sum_j x_j F_{ij})} \quad 3.3$$

Where F_{ij} is the mixture interaction factor which is correcting the contribution of each component. And it is calculated as:

$$F_{ij} = \frac{\left[1 + \left(\frac{\mu_i}{\mu_j} \right)^{1/2} \left(\frac{M_j}{M_i} \right)^{1/4} \right]^2}{\sqrt{8 \left(1 + \frac{M_i}{M_j} \right)}} \quad 3.4$$

The dynamic viscosity of the gases increase with temperature and the temperature dependent dynamic viscosity of a single gas can be calculated by a VDI polynomial fit [14]:

$$\mu_i(T) = A + BT + CT^2 + DT^3 + ET^4 \quad 3.5$$

Where A, B, C, D, E are empirical, substance specific fitting coefficients from VDI Heat Atlas tables.

Calculated viscosities at 1500K:

Gas	Viscosity (10^{-5} Pa s)
N ₂	5.41
CO ₂	5.20
CO	5,08
H ₂ O	4,28
O ₂	6,34

Table 3.5 : Calculated dynamic viscosities of species.

After substituting the values, the dynamic viscosity of the gas mixture is calculated as $6,92 \times 10^{-5} \text{ Pa s}$

Specific Heat Capacity of Gas Mixture

The specific heat capacity of the gases is also highly temperature dependent, and it determines the energy required to heat or cool the gas mixture. It is critical for calculating total heat transfer from off gas to scrap, scrap heating rates and thermal efficiency of the pre-heating process.

$$\frac{c_p}{R} = B + (C - B) \left(\frac{T}{A + T} \right)^2 \left[1 - \frac{A}{A + T} \left(D - E \frac{T}{A + T} + F \left(\frac{T}{A + T} \right)^2 + G \left(\frac{T}{A + T} \right)^3 \right) \right] \quad 3.6$$

Where C_p is the specific heat capacity of individual gases, R is the individual gas constant, T is the temperature and A, B, C, D, E, F, G values are gas specific parameters from VDI tables.

Gas	C_p at 1500 K (J/kg K)
N ₂	1245,79
CO ₂	1317,30
CO	1257,50
H ₂ O	2612,94
O ₂	1141,13

Table 3.6: Specific heat capacity of species

Then to calculate the specific heat of the gas mixture again the mole fraction weighted average method which was previously used to calculate mixture viscosity and molecular weight is used:

$$c_{p,mix} = \sum_{i=1}^N Y_i c_{p,i} \quad 3.7$$

After substituting the values specific heat capacity of gas mixture is calculated as 1330,20 (J/kg K)

Thermal Conductivity of the Gas Mixture

Thermal conductivity determines the conductive heat transfer, and it affects both the Prandtl number and the heat transfer coefficient from Nusselt number. The accuracy of the thermal conductivity calculation is also very critical therefore Wilke's rule for mixing is again used.

Thermal conductivity of individual gases is calculated using the VDI polynomial fit and gas specific parameters A, B, C, D, E from VDI tables.

$$\lambda_i = A + BT + C(T)^2 + D(T)^3 + E(T)^4 \quad 3.8$$

Gas	Thermal Conductivity W/mK
N ₂	0,0931
CO ₂	0,0904
CO	0.0870
H ₂ O	2,91

Table 3.7: Thermal conductivity of species.

Mixture interaction factor is also calculated and used to calculate the thermal conductivity of the gas mix. Then the gas mixture thermal conductivity is calculated by using Wilke's mixing rule

$$\lambda_{mix} = 0,25 \text{ W/mK} \quad 3.9$$

Molar Mass (M)	0,0293	kg/mol
Density (ρ)	0.2382	kg/m ³
Dynamic Viscosity (μ)	6,9228×10 ⁻⁵	Pa.s
Thermal Conductivity (λ)	0,25	W/mK
Specific Heat Capacity (c_p)	1330,21	J/kg · K

Table 3.8: Thermophysical properties of off gas.

Conservation Equations

The model for scrap preheating in the furnace shaft is based on the fundamental conservation laws of mass, momentum and energy [15].

Conservation of Mass

The conservation of mass for the gas phase flowing through the porous medium is as follows:

$$\frac{\partial(\varepsilon\rho)}{\partial t} + \nabla \cdot (\varepsilon\rho\mathbf{v}) = 0 \quad 3.10$$

Where ε is the porosity, ρ is the gas density, t is time and $\mathbf{v}$ is the superficial velocity.

The superficial velocity is defined as volumetric flow rate divided by the cross sectional area but treating the packed bed as an empty space. The porosity accounts for the volume fraction available for fluid flow.

Which states that the volumetric flow rate is conserved but, in this problem, the gas density varies with temperature change so the Ansys solves for the compressible flow.

Conservation of Momentum

The momentum conservation for single phase flow is governed by the Navier-Stokes equation:

$$\frac{\partial(\rho v)}{\partial t} + \nabla \cdot (\rho v v) = -\nabla p + \nabla \cdot \tau + \rho g \quad 3.11$$

Where p is the pressure, τ is the viscous stress tensor and g is the gravitation acceleration.

For flow through porous media, this equation needs to be volume averaged. Volume averaged conservation of momentum for single phase porous media is [16]:

$$\frac{\partial(\varepsilon \rho v)}{\partial t} + \nabla \cdot (\varepsilon \rho v v) = -\varepsilon \nabla p + \nabla \cdot (\varepsilon \tau) + \varepsilon \rho g + S_i \quad 3.12$$

In the volume-averaged formulation, the interaction between the gas phase and the solid scrap introduces an additional momentum sink term S_i , which represents the momentum exchange between the gas and the solid scrap due to the drag forces.

The flow in the fluid regions is governed by the standard Navier–Stokes, energy, and species transport equations. In the porous regions, the volume-averaged momentum and energy equations are solved, including Darcy–Forchheimer resistance and effective thermal conductivity to account for the packed bed.

Turbulence Treatment – RANS Approach

The high Reynolds number calculated previously indicates that the flow is turbulent. Since resolving all turbulent scales directly is computationally impractical, the Reynolds-Averaged Navier–Stokes (RANS) approach is adopted, where flow variables are decomposed into mean and fluctuating components.

RANS models provide the most economic approach for solving complex turbulent flows. These models introduce an Eddy-Viscosity term and two additional transport equations to calculate Reynolds stresses. RANS models are commonly used for various engineering applications, and they provide the required accuracy. The RANS models are not universal, so an appropriate model needs to be selected.

Due to its robustness, low computational cost and reasonable accuracy level, the $k-\varepsilon$ model, which is commonly used for EAF applications, is chosen for the CFD simulations in this study. While this model is known to be insensitive to boundary layer resolution and relies on wall functions, this limitation does not affect the objectives of the present study. Since the primary focus is the mixing behavior of the off gas in the bulk fluid region rather

than heat transfer to the walls, no wall refinement or low-Re corrections are applied to maintain computational efficiency.

The k-ε Turbulence Model in Ansys CFX

In the k-ε turbulence model k is the turbulence kinetic energy and it is defined as the variance of the velocity fluctuations and ε is the turbulence eddy dissipation which is the rate at which the velocity fluctuations dissipate.

The k-ε model introduces two new variables into the equations and the continuity equation becomes:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j} (\rho U_j) = 0 \quad 3.13$$

and the momentum equation:

$$\frac{\partial \rho U_i}{\partial t} + \frac{\partial}{\partial x_j} (\rho U_i U_j) = \frac{\partial p'}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu_{eff} \left(\frac{\partial \rho U_i}{\partial t} + \frac{\partial}{\partial x_j} \right) \right] + S_m \quad 3.14$$

Where S_M is the sum of body forces, μ_{eff} is the effective viscosity accounting for turbulence, and p' is the modified pressure.

The values of k and ε is calculated through the differential transport equations for turbulence kinetic energy and turbulence dissipation rate:

Turbulence kinetic energy (k):

$$\frac{\partial (\rho k)}{\partial t} + \frac{\partial}{\partial x_j} (\rho U_j k) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + (P_k - \rho \epsilon + P_{kb}) \quad 3.15$$

Dissipation rate:

$$\frac{\partial (\rho \epsilon)}{\partial t} + \frac{\partial}{\partial x_j} (\rho U_j \epsilon) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right] + \frac{\epsilon}{k} (C_{\epsilon 1} P_k - C_{\epsilon 2} \rho + C_{\epsilon 1} P_{\epsilon b}) \quad 3.16$$

Where $C_{\epsilon 1}$, $C_{\epsilon 2}$, σ_k , and σ_ϵ is the model constants defined by Ansys. P_{kb} and $P_{\epsilon b}$ represent the influence of the buoyancy forces and P_k is the turbulence production due to viscous forces which is calculated by ANSYS.

The Boussinesq hypothesis is adopted to close the RANS equations, relating the Reynolds stresses to the mean strain rate through an eddy viscosity. Although this approach assumes isotropic turbulence, it is suitable for the present study, which focuses on bulk gas mixing with high Reynolds numbers in a large fluid region rather than detailed near-wall turbulence structures. The Boussinesq model that is solved by ANSYS is:

$$P_{kb} = \frac{\mu_t}{\rho \sigma_\rho} \rho \beta g_i \frac{\partial T}{\partial x_i} \quad 3.17$$

Porous Media Momentum Sink

The momentum source term S_i in the RANS equations represent drag forces exerted by solid scrap on the gas flow, calculated using the Ergun equation:

$$S_i = - \left(\frac{\mu}{\alpha} v_i + C_2 \frac{\rho}{2} |v| v_i \right) \quad 3.18$$

Where S_i is the source term of the momentum equation in the i direction, v_i is the velocity in i direction and v is the magnitude of the velocity. The resistance coefficients calculated from this correlation equation are entered directly into porous zone in Ansys.

Permeability:

$$\alpha = \frac{d_p^2 \varepsilon^3}{150(1 - \varepsilon)^2} \quad 3.19$$

Inertial resistance:

$$C_2 = 3.5 \frac{(1 - \varepsilon)}{d_p \varepsilon^3} \quad 3.20$$

These values depend on the particle size and porosity but remain constant for any mass flow rate and Ansys calculates the resulting pressure drop and velocity based on given boundary conditions of mass flow rate at the inlet.

Conservation of Energy

For the scrap preheating problem, a thermal non equilibrium model is required because the gas and solid phases exchange heat on a finite timescale that is the preheating residence time. Due to large temperature difference and short preheating time, the gas and solid cannot be assumed to be in thermal equilibrium. Therefore, separate energy equations are required for each phase.

Energy equation for the fluid phase in a porous medium for thermal non-equilibrium [15, 18]:

$$(\varepsilon \rho c_{p,g}) \left(\frac{\partial T_g}{\partial t} + v \cdot \nabla T_g \right) = \nabla \cdot (\varepsilon \lambda_g \nabla T_g) + h f_s \alpha (T_s - T_g) + S_{chem} \quad 3.21$$

Where T_g and T_s are the gas and solid temperatures, $C_{p,g}$ is the specific heat of the gas, λ_g is the thermal conductivity of gas, $h f_s$ is the gas-solid heat transfer coefficient and a is the interfacial area density. S_{chem} is the heat addition from CO oxidation.

Solid Phase

$$(1 - \varepsilon) \rho c_{p,s} \left(\frac{\partial T_s}{\partial t} \right) = \nabla \cdot [(1 - \varepsilon) k_s \nabla T_s] + h f_s \alpha (T_g - T_s) \quad 3.22$$

Left side of the equation gives how fast the thermal energy is accumulated with volume of the scrap, density, specific heat and temperature.

First term of the right hand side of the equation gives the heat conduction within the scrap pieces with ∇T_s being the temperature gradient in the scrap, $k_s \nabla T_s$ heat flux by Fouriers law, $(1 - \varepsilon)$ again represents the ratio of volume that the scrap occupies and $\nabla \cdot [(1 - \varepsilon) k_s \nabla T_s]$ being the divergence in heat flux.

The second term of the right-hand side $h f_s \alpha (T_g - T_s)$ captures the convective heat transfer from hot gas to cold scrap. This term has the same magnitude but opposite sign with the second term of the right-hand side in gas phase because the solid scrap is taking the energy and gas is giving it to the solid.

As the gas cools down or the scrap heats up, the temperature difference decreases, and it causes the heat transfer rate to drop. This effect is captured by CFD simulation by solving the energy equation for two phases simultaneously.

Heat Transfer in Packed Beds

The scrap inside the shaft absorbs heat through convection radiation and conduction within the scrap. The industrial steel scrap consists of irregularly shaped particles with non-uniform geometries and sizes, and it is not possible to model these variations for CFD applications. The CFD tool Ansys that is used for this application assumes a homogenous porous region where the scrap is and it requires the porosity, resistance coefficients and heat transfer coefficients as inputs for simulations. These inputs are dependent on particle size and that makes the correct particle size selection important.

Mandal (2010) conducted experimental investigations on transient heating of irregularly shaped scrap by hot gases, examining scrap pieces with characteristic dimensions ranging from 63 mm to 195 mm. His experimental work demonstrated that scrap beds with particles in this size range exhibit porosity values between 0.86 and 0.95, depending on packing arrangement and particle shape distribution. These measurements provide guidance for selecting representative particle diameters for CFD modeling of scrap preheating.

To determine the required inputs for CFD a particle diameter of 0.1m was chosen as the baseline value within the range provided by Mandal (2010). Also, a sensitivity study for different particle sizes from 0.05m to 0.3m has been conducted to understand the effects of particle size in scrap preheating. Based on the given particle diameter different heat transfer phenomena for this scrap preheating problem is investigated.[19]

Convective Heat Transfer

The Gunn Correlation

One of the most widely used correlations for gas-solid convective heat transfer in packed beds is the Gunn correlation (Gunn, 1978). The Gunn correlation is applied to calculate the heat transfer coefficient in this study since it accounts for the porosity which directly influence the turbulence intensity and effective gas-solid contact area, therefore the convective heat transfer coefficient. (Hassan 2025) The first term of the correlation accounts for the

conduction and the second term accounts for the convection. At high Re conditions convection dominates. Using the Gunn correlation the Nusselt number is calculated [20, 21]:

$$Nu = (7 - 10\varepsilon + 5\varepsilon^2)(1 + 0.7Re_p^{0.2}Pr^{1/3}) + (1.33 - 2.4\varepsilon + 1.2\varepsilon^2)Re_p^{0.7}Pr^{1/3} \quad 3.23$$

Where Nu is the Nusselt number, h is the convective heat transfer coefficient [W/(m²·K)] and Pr is the Prandtl number.

Calculation for Scrap Preheating Conditions

For the operating conditions in the present study, the relevant parameters at gas temperature of 1500 K are:

Gas thermal conductivity: (λ_g)	0.25	W/(m·K)
Gas dynamic viscosity: (μ)	6.92×10^{-5}	Pa·s
Gas specific heat: (c_p)	1330	J/(kg·K)
Gas density: (ρ)	0.23	kg/m ³
Particle diameter: (d_p)	0.1	m
Porosity: (ε)	0.85	-

Table 3.9: Relevant parameters to calculate the heat transfer coefficient.

The Prandtl number, which is constant for all flow rates is calculated as:

$$Pr = \frac{\mu c_p}{k_g} \quad 3.24$$

The convective heat transfer coefficient (h_{fs}) is then calculated using the Nusselt number:

$$h_{fs} = \frac{Nu \cdot k_g}{d_p} \quad 3.25$$

The interfacial area density for spherical particles in a packed bed is

$$\left(A_{fs} = \frac{6(1 - \varepsilon)}{d_p} \right) \quad 3.26$$

To see the effects of mass flow rates, the values depending on the mass flow rate in heat transfer from gas to scrap is calculated for different mass flow rates.

	5 kg/s	25 kg/s	45 kg/s
Prandtl Number	0,368	0,368	0,368
Nusselt Number	14,295	33,467	47,489
HTC (W/(m ² K))	35,766	83,73452	118,8199

Table 3.10: Calculated heat transfer coefficients, Nusselt and Prandtl numbers

Table 3.10 shows the calculated heat transfer coefficients, Nusselt and Prandtl numbers calculated at three different mass flow rates of 5kg/s, 25kg/s and 45kg/s

On the table heat transfer coefficients for conditions that will be simulated are given.

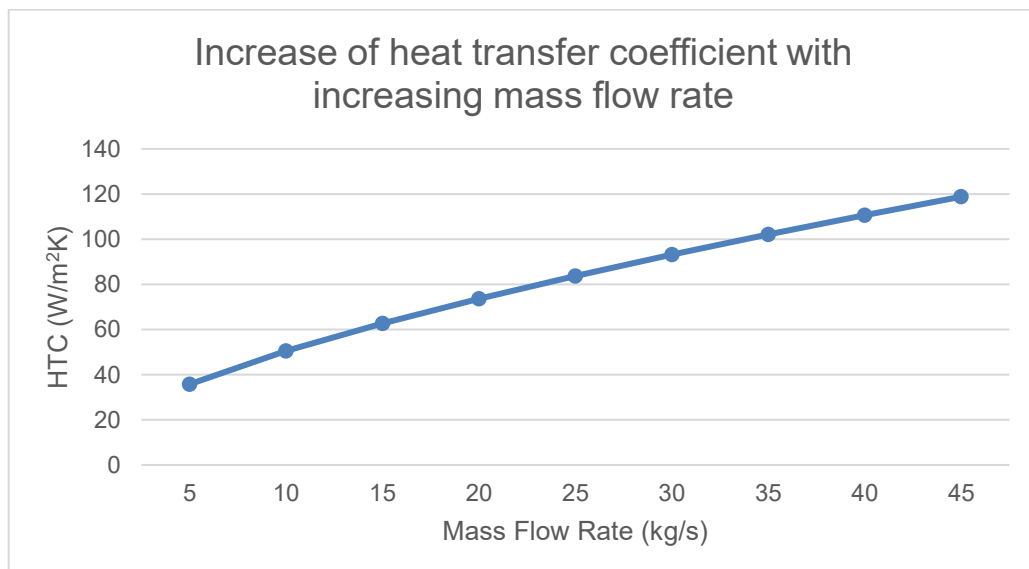


Figure 3.3 Plot of heat transfer coefficient with increasing mass flow rate

Figure 3.3 shows how heat transfer coefficient increases linearly with mass flow rate. The coefficient rises from 35.7 watts per square meter kelvin at 5 kilograms per second to 118.8 watts per square meter kelvin at 45 kilograms per second

In this problem where the scrap pieces are assumed to be spherical, the heat transfer coefficient between gas and solid decreases mainly due to larger particles having lower specific surface area, meaning less interfacial surface area per unit volume for heat exchange and the Nusselt number increases with particle diameter. The heat transfer coefficient is inversely proportional to particle diameter when calculated from the Nusselt number.

This increase in the heat transfer coefficient with the decreasing particle size explains the plant observations of melting scrap pieces at the bottom of the scrap bed. Expected average temperature of the scrap bed is much lower than the melting temperatures of steel but the particles with high surface area to volume ratio such as sheet metal or small pieces can reach to much higher temperatures than the average bulk temperature of the scrap bed.

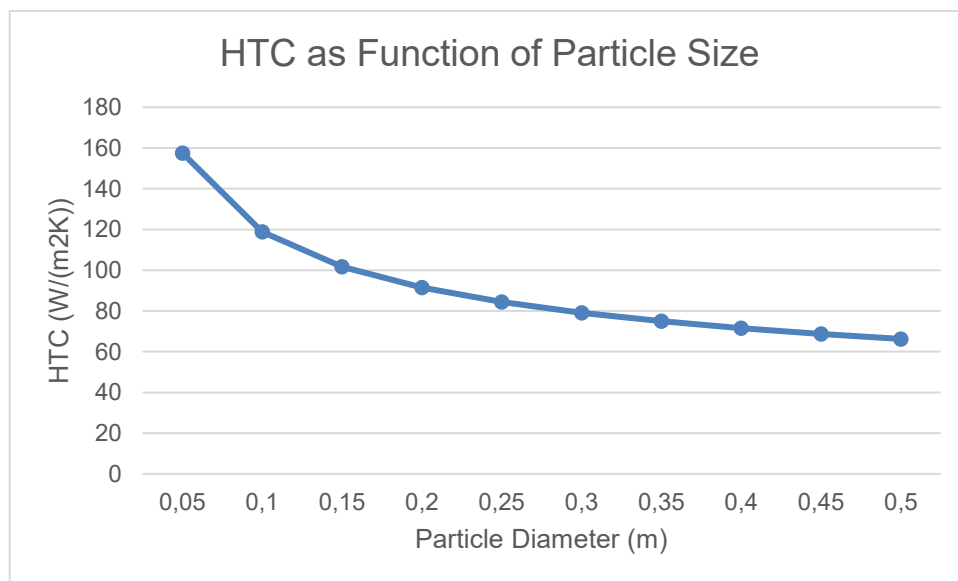


Figure 3.4 Relationship of HTC and particle size

Figure 3.4 shows the decrease in HTC as the particle diameter increases at 45kg/s off gas mass flow rate.

Conductive Heat Transfer

Based on the heat transfer coefficients that were calculated previously, conduction can be investigated. When the heat conduction within the solid is much faster than the rate of convective heat transfer from the fluid, the individual solid particles can be assumed to have uniform internal temperature. This approximation is known as lumped capacitance, and it simplifies the solid energy equation by eliminating internal temperature gradients. To assess the validity of this approach, a dimensionless parameter named Biot number (Bi) needs to be determined, and it is defined as follows [22]:

$$Bi = \frac{h_{fs} \times L_c}{k_s} \quad 3.27$$

Where h_{fs} is the convective heat transfer coefficient between gas and scrap [$\text{W}/(\text{m}^2.\text{K})$], L_c is the characteristic length of the solid particle [m] and k_s is the thermal conductivity of steel [$\text{W}/(\text{m.K})$]

For a sphere, the characteristic length is defined as the ratio of volume to surface area:

$$L_c = \frac{V_p}{A_p} = \frac{\frac{\pi d_p^3}{6}}{\pi d_p^2} = \frac{d_p}{6} \quad 3.28$$

This length scale represents the average distance heat must travel by conduction from the particle surface to its center.

The biot number represents the ratio of internal conductive thermal resistance to external convective thermal resistance:

$$Bi = \frac{R_{conduction}}{R_{convection}} = \frac{L_c/k_s}{1/h_{fs}} \quad 3.29$$

Biot number greater than one indicates significant temperature gradients and the lumped capacitance approximation is generally considered valid when $Bi < 0.1$.

Particle Diameter d_p (m)	Characteristic Length L_c (m)	Heat Transfer Coefficient h_{fs} (W/m ² ·K)	Biot Number Bi
0,05	0,008	136,63	0,025
0,1	0,016	102,09	0,037
0,15	0,025	87,05	0,048
0,2	0,033	78,07	0,057
0,25	0,041	71,89	0,066
0,3	0,05	67,29	0,074

Table 3.11: Calculated Biot numbers

To ensure the lumped capacitance assumption remains valid for all particle sizes considered in this study, Biot numbers are calculated for the full range of particle diameters considering steel thermal conductivity $k_s = 45$ W/ (m.K). Table 3.11 summarizes the resulting Biot numbers.

Biot number for each particle size indicates that the lumped capacitance assumption is valid across entire range and even at the largest particle size ($d_p = 0.30$ m), the Biot number ($Bi = 0.074$) remains well below the validity limit (0.1), providing confidence in the model assumptions. Based on these Biot number calculations, internal conduction modeling within the individual scrap particles is not necessary for present study.

Transient Scrap Preheating

The CFD model solves the conservation equations for mass, momentum and energy for both the gas and the solid phases. The scrap preheating process works as a batch system where each batch of scrap is heated in the shaft for approximately 5 to 7 minutes.

Thus, a transient approach is necessary to capture the time dependent preheating process. This section first explains why transient simulation is necessary, then presents the governing equations of this application.

Justification For Transient Approach

The scrap preheating process in the EAF shafts that are investigated in this paper is not a continuous process. The scrap is being charged inside the shaft batch by batch and before they are discharged into the furnace, they spend approximately 5 to 7 minutes, during which it is being exposed to hot gas and partially heats.

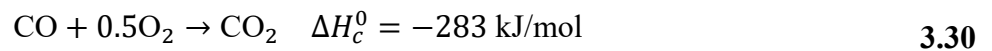
Steady state simulations were performed previously to assess whether quasi steady state conditions could be assumed. However, these simulations resulted in scrap being overheated and outlet temperatures of 1300-1400K. This indicates that the steady-state solutions assume infinite time, which allows complete thermal equilibrium between gas and scrap.

7-8 minutes of preheating time is not sufficient for thermal equilibrium between gas and scrap. Therefore, a transient simulation approach is required to make realistic predictions of scrap and gas temperatures.

Carbon Monoxide Combustion

Reaction Mechanism

CO oxidation in the EAF shaft is modeled using a single-step global mechanism:



This approach is widely employed in EAF combustion CFD studies as it captures the overall heat release and reaction rate while maintaining computational efficiency. Single-step mechanisms are frequently used for industrial furnace simulations where the primary interest is overall energy recovery rather than detailed flame structure. [5]

Stoichiometric Analysis

Inlet Conditions

The off gas is considered to have no oxygen but the pressure drop across the shaft causes air infiltration into the vessel. First the required oxygen to fully combust the CO is calculated and Ansys simulations were made using the required oxygen amount for complete combustion then the oxygen that is available is calculated based on the area of the gas that air infiltrates into the system and another run was made so the results can be compared.

Gas Composition	Mass Basis (%)	Molar Basis (%)
N ₂	60	62.82
CO ₂	20	15.71
CO	15	13.33
H ₂ O	5	8.15

Table 3.12: Mass and molar fractions of individual species.

Calculation of CO Flow Rates

The molar flow rate of the off gas is calculated from the mass flow rate and average molecular weight.

$$\dot{n}_{total} = \frac{\dot{m}_{total}}{M_{total}} \quad 3.31$$

The CO content in the off gas, is calculated as:

$$\dot{n}_{CO} = x_{CO} \times \dot{n}_{total} \quad 3.32$$

CO mass flow rate where mole fraction is 0,157:

$$\dot{m}_{CO} = 0.157 \times \dot{m}_{total} \quad 3.33$$

Flow rates of CO at different inlet flow conditions

Total Flow (kg/s)	Total Molar Flow (mol/s)	CO Molar Flow (mol/s)	CO Mass Flow (kg/s)
5	170,48	26,77	0,75
15	511,44	80,32	2,25
25	852,40	133,88	3,75
35	1193,36	187,43	5,25
45	1534,32	240,98	6,75

Table 3.13: Calculated total molar flow rate, CO molar and mass flow rates

Oxygen Requirements

From the stoichiometry of the reaction $\text{CO} + 0.5\text{O}_2 \rightarrow \text{CO}_2$, each mole of CO requires 0.5 moles of O_2 for complete combustion. The calculation of required oxygen is as follows:

$$\dot{n}_{\text{O}_2, \text{req}} = 0.5 \times \dot{n}_{\text{CO}} \quad 3.34$$

$$\dot{m}_{\text{O}_2, \text{req}} = \dot{n}_{\text{O}_2, \text{req}} \times M_{\text{O}_2} \quad 3.35$$

Since the inlet gas contains no oxygen, supplemental oxygen injection is modeled for post-combustion in the shaft. To ensure complete combustion and account for mixing imperfections in the shaft, a 20% excess oxygen is supplied:

$$m_{\text{O}_2, \text{inject}} = 1.2 \times m_{\text{O}_2, \text{req}} \quad 3.36$$

Base Flow (kg/s)	O ₂ Required (kg/s)	O ₂ Injected (kg/s)	Total Flow (kg/s)
5	0,42	0,51	5,51
15	1,28	1,54	16,54
25	2,14	2,57	27,57
35	2,99	3,59	38,59
45	3,85	4,62	49,62

Table 3.14: total mass flow rate with oxygen.

The total gas mass flow rate entering the shaft with oxygen injection becomes approximately 10% higher than the baseline flow without combustion.

Calculating Air Infiltration

Under normal operating conditions the oxygen that is required for the combustion comes from the air that is leaking into the vessel through the gaps around the furnace and the amount of air that infiltrates into the vessel is dependent on the pressure difference between the outlet of the shaft and the vessel. The pressure difference is monitored constantly during operation and the total area of gaps is known so the mass flow rate of leaking air can be calculated using Orifice equation [25]:

$$\dot{m}_{\text{air}, \text{leak}} = C_d A \sqrt{2\rho \Delta p} \quad 3.37$$

Where C_d is the discharge coefficient (0.6 for sharp edged furnace openings), A is the leakage area, ρ is the air density and Δp is the pressure difference.

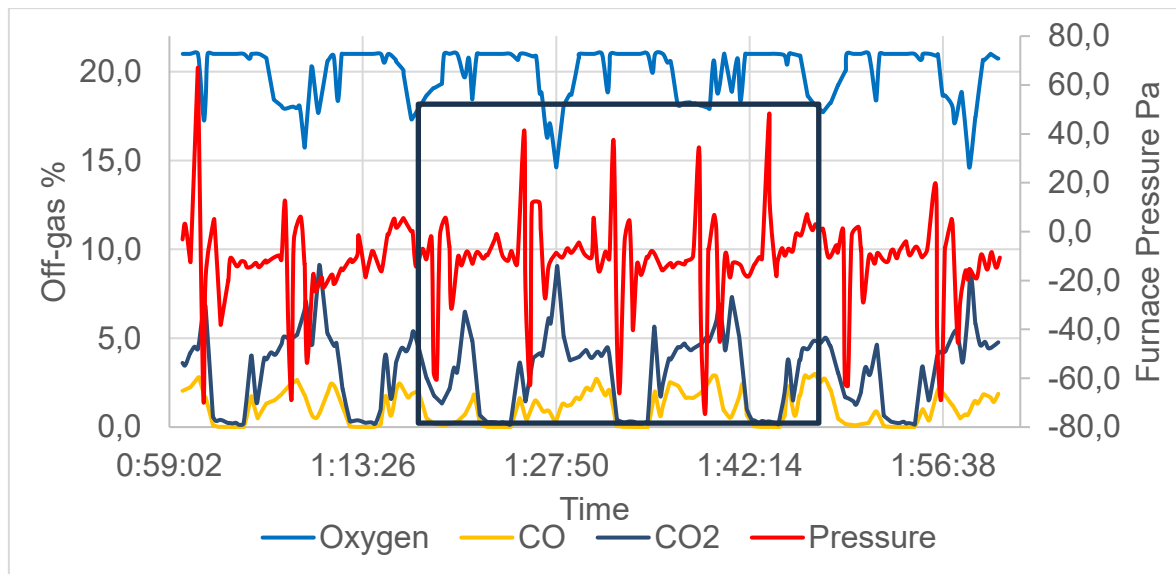


Figure 3.5 Pressure and Gas Composition Measurements

In the figure 3.5 plot of typical pressure drop across the shaft during operation can be seen. The peaks in pressure measurement show the moment when the scrap is discharged into the furnace from the shaft which causes overpressure. Whereas the drops are measured when the flap is opened for the next batch charging and the seal is broken. This drop is then followed by a smaller peak which corresponds to the moment where the scrap is charged into the shaft.

In the figure 3.5 the change in species concentration during the operation is also shown. The increase in furnace pressure limits the false air flow into the system therefore the CO and CO₂ concentrations rise, and the oxygen levels drop.

In the actual EAF operation, air enters the system through gaps near the roof driven by the pressure difference between the shaft outlet and atmospheric pressure. This air mixes with CO rich off gas throughout the furnace freeboard and combusts before entering the shaft inlet and most of the resulting heat is absorbed by the water-cooled panels in the furnace freeboard.

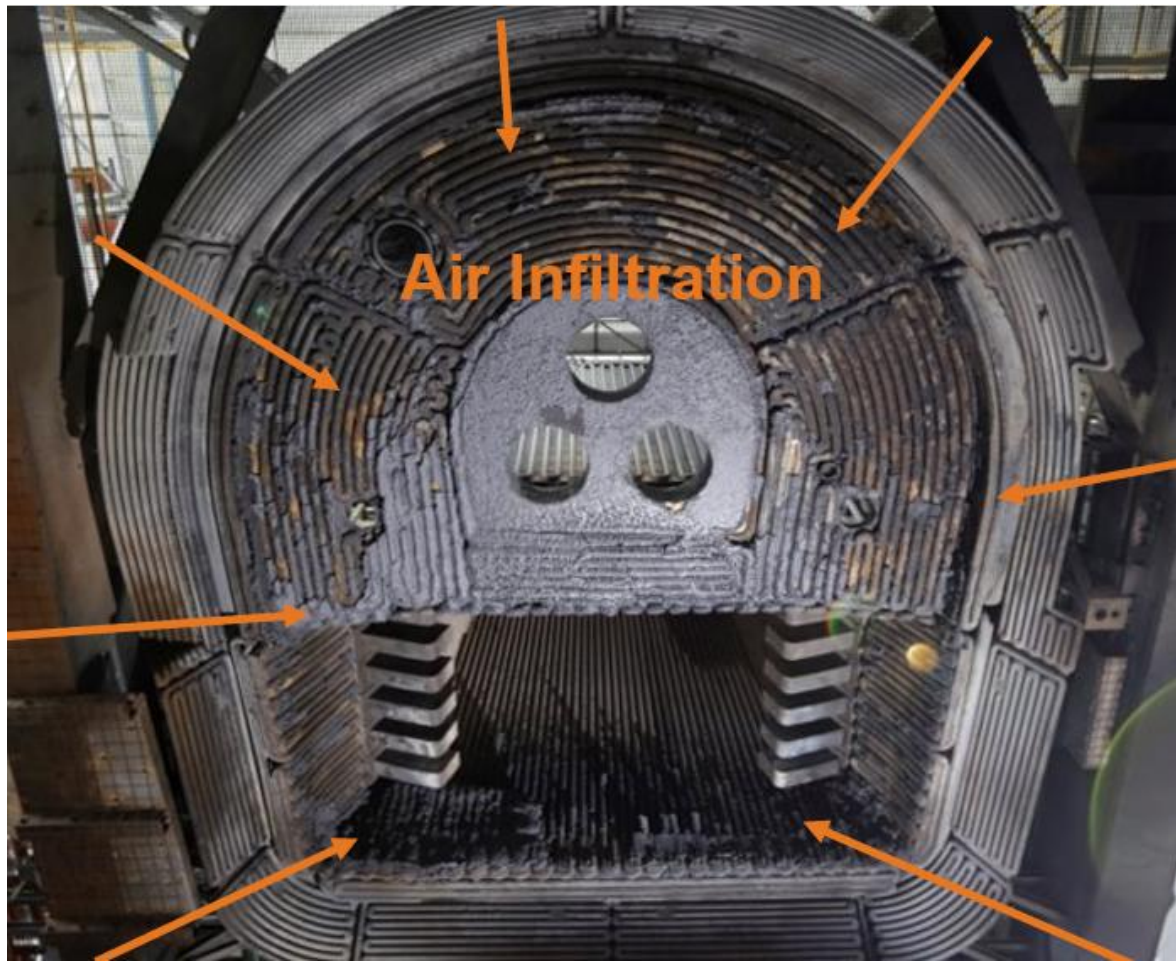


Figure 3.6: Furnace lid from the bottom

Figure 3.6 Shows the furnace lid from the bottom. Air infiltrates into the furnace through the gaps where the lid meets the upper shell. Shaft inlet can be seen at the bottom of the figure.

The focus of the present CFD model is the furnace shaft therefore it does not include complete furnace geometry and having a more complex geometry increases the computational costs. To represent this air leakage into the system, discrete air inlets were defined near shaft inlet. This approach assumes all CO combustion takes place where all heat resulted from the combustion is available for scrap preheating. Therefore, the simulation of CO combustion and scrap heating rates may represent an upper bound on achievable performance.

Turbulence-Chemistry Interaction (Eddy Dissipation Model)

The interaction between turbulence and chemical reactions is captured using the Eddy Dissipation Model (EDM), originally developed by Magnussen and Hjertager (1977). The EDM assumes that in turbulent flows, the reaction rate is controlled by the rate at which reactants are mixed at the molecular level through turbulent eddies [26].

Ansys provides a turbulence-chemistry interaction model based on the work of Magnussen and Hjertager. The net production rate of species due to the reaction is determined by the smaller value of the two expressions [30]:

$$R_{i,r} = v_{i,r} M_{w,i} A \rho \frac{\epsilon}{k} \cdot \min \left(\frac{Y_R}{v'_{R,r} M_{w,R}} \right) \quad 3.38$$

$$R_{i,r} = v_{i,r} M_{w,i} A B \rho \frac{\epsilon}{k} \left(\frac{\sum_P Y_P}{\sum_j^N v''_{j,r} M_{w,j}} \right) \quad 3.393$$

where:

- $v'_{i,r}$ = stoichiometric coefficient for reactant i in reaction r
- $v''_{j,r}$ = stoichiometric coefficient for product j in reaction r
- $M_{w,i}$ and $M_{w,j}$ = molecular weight of species i and j [kg/kmol]
- A = empirical constant (default value 4.0)
- ρ = density [kg/m³]
- ϵ = turbulent dissipation rate [m²/s³]
- k = turbulent kinetic energy [m²/s²]
- Y_R = mass fraction of a particular reactant
- Y_P = mass fraction of any product species
- N = number of chemical species

The model constant A controls the reaction rate and has a default value of 4.0, which has been validated for a wide range of combustion applications including EAF off gas systems [27].

Damköhler Number Analysis

To verify that the EDM is appropriate for this application Damköhler number, which assess the relative importance of chemical kinetics versus turbulent mixing, is calculated

The Damköhler number compares reaction and transport time scales:

$$Da = \frac{\tau_{\text{res}}}{\tau_{\text{chem}}} \quad 3.40$$

Where chemical time scale is calculated by dividing molar concentration of CO by the reaction rate:

$$\tau_{\text{chem}} = \frac{[CO]}{R_{CO}} \quad 3.41$$

Chemical Time Scale

The chemical time scale is calculated using the Dryer-Glassman kinetics [28] for CO oxidation in the presence of H₂O:

$$R_{CO} = C \times \exp\left(\frac{-E_a}{RT}\right) \times [CO]^a \times [H_2O]^b \times [O_2]^c \quad 3.42$$

where:

- $C = 3,9 \times 10^{14}$
- $E_a = 40 \text{ kcal/mole}$
- $a = 1.0, b = 0.5, c = 0.25$

The Dryer-Glassman mechanism is only used as a reference to calculate the characteristic time scale for the Damköhler number calculation which validates the EDM assumption of fast chemistry relative to mixing. The actual CFD simulations employ EDM which does not require input of detailed kinetic parameters.

At inlet conditions ($T = 1500 \text{ K}$) with representative species concentrations:

The reaction rate becomes:

$$[CO] = 0,15 \times \left(\frac{P}{RT} \right) = 1.183 \times 10^{-6} \text{ mol/cm}^3 \quad 3.44$$

$$[O_2] = 5.915 \times 10^{-7} \text{ mol/cm}^3 \quad 3.45$$

$$[H_2O] = 6.134 \times 10^{-7} \text{ mol/cm}^3 \quad 3.46$$

$$R_{CO} = 1.484 \times 10^{-2} \text{ mol/(cm}^3 \cdot \text{s)} \quad 3.43$$

This calculation uses a simplified approach assuming uniform concentrations. Actual concentrations can vary across the CFD domain.

Calculated value of $7,67 \times 10^{-5}$ for the chemical time scale is in alignment with the literature. From the work of Wartha et al. [29] the chemical time scales calculated through different definitions for the simplified one step carbon monoxide oxidation is between 10^{-7} and 10^{-1} .

Gas Residence Time

The residence time is the average time gas spends in the shaft with a total length of 10 meters, and the superficial velocity is calculated for each mass flow rate. Then it used to calculate Damköhler number:

$$\tau_{\text{res}} = \frac{L}{v_s} \quad 3.47$$

Where v_s is the superficial velocity calculated from the final mass flow rate with oxygen addition and L is the shaft length.

Initial Mass Flow Rate	Mass Flow Rate with Oxygen Addition	Superficial Velocity	Residence Time	Damköhler Number
5	5,51	1,54	6,48	84459
10	11,02	3,08	3,24	42229
15	16,54	4,62	2,16	28153
20	22,05	6,17	1,62	21114
25	27,57	7,71	1,29	16891
30	33,08	9,25	1,08	14076
35	38,59	10,79	0,92	12065
40	44,11	12,34	0,81	10557
45	49,62	13,88	0,72	9384

Table 3.15: Damköhler numbers

Table 3.15 shows the Damköhler numbers, residence times and superficial velocities with corresponding mass flow rates with oxygen addition. Damköhler number calculation is based on the mass flow rate with oxygen addition and initial mass flow rates are listed for reference.

Damköhler number that is significantly greater than 1 indicates that the chemistry is significantly faster than transport and combustion rate is mixing limited, not kinetically limited.

Therefore, turbulent mixing controls overall combustion rate. Eddy Dissipation Model is appropriate which assumes fast chemistry. Combustion zone will be narrow and intense near inlet point and the temperature effects on kinetics are secondary to mixing effects

Species Transport

Dominant Transport Mechanism – Peclet Number

The Peclet number determines the relative importance of convective transport to diffusive transport:

$$Pe = \frac{v_s \times d_p}{D_{AB}} \quad 3.48$$

where v_s is the superficial velocity, d_p is the particle diameter, and D_{AB} is the binary molecular diffusion coefficient for CO-N₂.

The binary diffusion coefficient at high temperature is calculated using the Chapman-Enskog theory with Lennard-Jones potential parameters and the Neufeld collision integral correlation [13]. For CO-N₂ at 1500K and 1 atm, D_{AB} is approximately $2.5 \times 10^{-4} \text{ m}^2/\text{s}$

$\dot{m}(\text{kg/s})$	$\dot{m} \text{ with } \text{O}_2 (\text{kg/s})$	$V_s (\text{m/s}) / D_p (\text{m})$	0,05	0,1	0,15	0,2	0,25	0,3
5	5,51	1,54	246	493	739	986	1232	1479
10	11,02	3,08	493	986	1479	1972	2465	2958
15	16,54	4,62	739	1479	2218	2958	3697	4437
20	22,05	6,17	986	1972	2958	3944	4930	5916
25	27,57	7,71	1232	2465	3697	4930	6162	7395
30	33,08	9,25	1479	2958	4437	5916	7395	8874
35	38,59	10,79	1725	3451	5176	6902	8627	10353
40	44,11	12,34	1972	3944	5916	7888	9860	11832
45	49,62	13,88	2218	4437	6655	8874	11093	13311

Table 3.16: Peclet Numbers

Table 3.16 shows the Peclet numbers for each particle size and mass flow rate with oxygen addition. Initial mass flow rates are listed for reference.

The Peclet number ranges from approximately 200 to 13000 and $Pe > 1$ indicates that the species transport is heavily dominated by convection and turbulent mixing. Molecular diffusion contribution is negligible, where turbulent diffusivity is:

$$D_{turb} = \frac{\mu_t}{\rho Sc_t} \quad 3.49$$

where $Sc_t = 0.7$ is the turbulent Schmidt number for gaseous species (ANSYS default).

This turbulent diffusivity appears in the Eddy Dissipation Model, which calculates the mixing-limited reaction rate.

The combined analysis of high Peclet and high Damköhler numbers validates the modeling approach: species transport is convection-dominated, and combustion is mixing-limited.

4. Results and Discussion

Scrap Preheating Without CO Combustion Baseline Characterization

Effect of Off gas Mass Flow Rate on Scrap Preheating

Sizes and shapes of scrap particles vary within the batch, and the mass flow rate of the off gas can vary throughout the process.

Simulations were conducted at varying off gas flow rates without CO combustion to establish baseline heat transfer behavior. The scrap bed was modeled as a porous domain with uniform spherical particles of 0.1m diameter. Operating conditions are summarized in Table 4.1.

Case m (kg/s)	Outlet Gas Temperature (K)	Average Scrap Temperature (K)	ΔT_{scrap} (K)	h_{fs} (W/m ² K)
5	315	342	42	35,76
10	366	382	82	50,45
15	424	419	119	62,71
20	470	454	154	73,66
25	526	485	185	83,73
30	569	514	214	93,16
35	616	543	243	102,01
40	650	571	271	110,62
45	680	598	298	118,81

Table 4.1: Scrap temperatures in 5 minutes

Table 4.1 Shows the scrap temperatures after 300 seconds (5 minutes) of preheating for each mass flow rate with corresponding Nusselt numbers and heat transfer coefficients. Scrap outlet temperatures reached 300-600K. The volume-averaged scrap temperatures show more uniform distribution than expected from plant observations of localized scrap melting.

The off gas temperatures at the shaft outlet is also presented in table 4.1 showing that with increasing mass flow rate, the outlet gas temperature also increases.

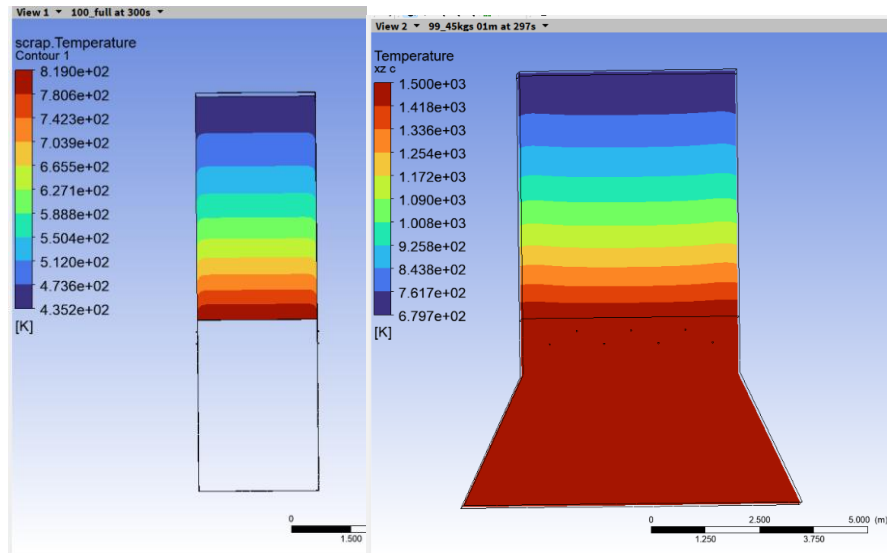


Figure 4.1: Temperature gradient of scrap (left) and gas (right)

Figure 4.1 Shows the temperature gradient of scrap (left) and gas (right) for a mass flow rate of 45kg/s and particle size of 0.1m.

Effect of Particle Size on Heat Transfer Characteristics

To investigate the apparent discrepancy between simulation predictions and plant observations, a parametric study varying particle diameter from 0.1m to 0.3 m was conducted. Results are shown in Table 4.2.

dp (m)	T _{out} (K)	T _{scrap,avg} (K)	ΔT _{scrap} (K)	h _{fs} (W/m ² K)
0.10	424	419	119	118,81
0.15	690	390	90	101,75
0.20	844	372	72	91,48
0.25	962	359	59	84,40
0.30	1058	348	48	79,10

Table 4.2 Scrap Temperatures After 5 Minutes of Preheating

The heat transfer coefficient increased dramatically as particle size decreased, following the expected relationship from the Gunn correlation. For particles of 0.1 m diameter, the

heat transfer coefficient was approximately 2 times higher than for 0.3 m chunks. This sensitivity explains the observed discrepancy between CFD predictions and plant reports of scrap melting. Large scrap pieces with low surface area-to-volume ratio heat slowly and remain below melting temperature, while sheet metals or small particles with high surface area-to-volume ratio can rapidly reach temperatures where they can start melting and dripping. The uniform sphere assumption in the CFD model averages these extremes, predicting moderate temperatures throughout the bed that are consistent with bulk scrap behavior while not capturing the localized hot spots where thin sections melt and drip.

CFD simulations of a packed bed of scrap require a constant average particle size. Uniformly distributed, same sized spherical particles are default therefore the effects of different shapes and sizes in a single batch cannot be captured.

In alignment with literature in packed scrap bed studies, the particle diameter of 0.1m is chosen as baseline.

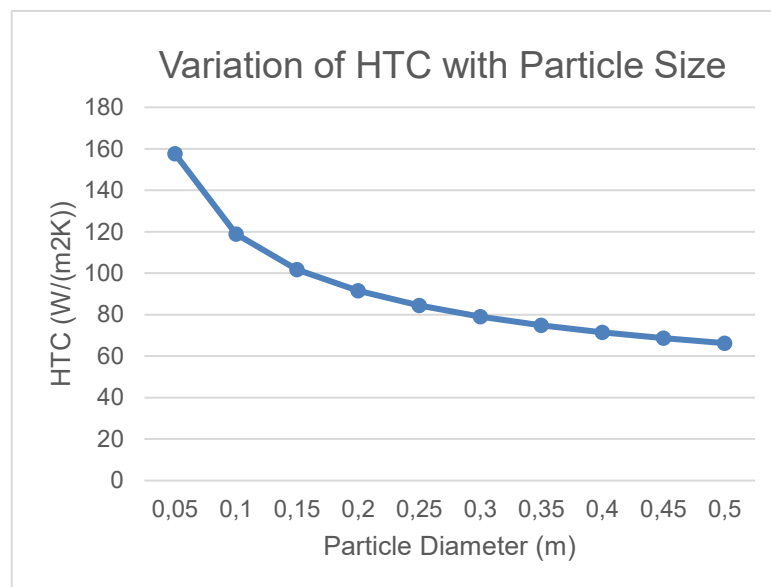


Figure 4.2: Variation of heat transfer coefficient (h_{fs}) with particle size.

It is important to note that particles with higher surface area to volume ratio such as sheets, rebars and so on have higher heat transfer coefficients so they can reach much higher temperatures. But the volume averaged temperature of the bulk scrap would remain the same.

First simulation group, group A, representing the state where the scrap is discharged into the furnace. In this state the flap is open so the suction is broken. In this state a sudden

overpressure occurs inside the furnace and the off gas leaves the system from various gaps rather than being sucked through the shaft.

A	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
A1	35	15	open	0	no	27
A2	40	15	open	0	no	27
A3	45	15	open	0	no	27
A4	35	15	closed	-1000	no	27
A5	40	15	closed	-1000	no	27
A6	45	15	closed	-1000	no	27
A7	35	15	closed	-1000	yes	27
A8	40	15	closed	-1000	yes	27
A9	45	15	closed	-1000	yes	27

Table 4.3: Group A Parameters

When CO rich, hot off gas leaves the system, it contacts air and causes flames and explosions as it was shown in section 2. This effect is captured by maintaining the baseline off gas mass flow rate at 35, 40 and 45 kg/s and outlet pressure and ambient pressure equal.

A	O2 Mass Fraction	O2 Mass Flow(kg/s)	CO Mass Fraction	CO Mass Flow (kg/s)	CO2 Mass Fraction	False Air (kg/s)
A1	0	0	0,15		0,2	-1
A2	0	0	0,15		0,2	-1,13
A3	0	0	0,15		0,2	-1,27
A4	0,058	3,81	0,00011	0,0077	0,23	29,83
A5	0,048	3,41	6,89E-05	0,0048	0,24	29,73
A6	4,00E-02	3	0,0009	0,06	0,26	29,63

Table 4.4: Group A Results

Variation of mass flow rate did not change the leaking off gas mass flow rate, which are shown in the table at False Air column but with negative sign indicating that the off gas is flowing out of the system where normally the false air infiltrates the system, significantly. Actual amount of off gas that leaves the system in reality based on the overpressure and the amount of available off gas inside the furnace but the effect of furnace overpressure is captured clearly.

When the cases of A4, A5, A6 and A7, A8, A9 are compared, the chimney effect of the empty shaft with open flap is seen. The CFD does not automatically capture the chimney effect, and the flap is not modeled directly so saying that the flap is opened or closed here is only a representation. Considering the same suction pressure for all cases and removing the scrap for first three cases and adding it for the other three is the closest representation.

A	O2 mass fraction	O2 mass flow (kg/s)	CO mass fraction	CO mass flow (kg/s)	CO2 mass fraction	False Air (kg/s)
A7	0,002	1,14	0,133	0,687	0,27	16,56
A8	0,001	0,55	0,031	1,69	0,27	13,4
A9	0	0,08	0,057	3,11	0,27	9,59

Table 4.5: Group A results

When the false air mass flow rates are compared it is seen that almost 30kg/s of false air infiltrates the system in cases without scrap for all off gas mass flow rates but when the scrap is present almost half of the false air at 16kg/s infiltrates the system and decreasing to approximately 10 kg/s at the worst case. This is in alignment with plant observations where the flames can be seen at the hood area while charging and discharging the scrap, as it was discussed in section 2.

B	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
B1	35	15	closed	-600	yes	27
B2	35	15	closed	-800	yes	27
B3	35	15	closed	-1000	yes	27
B4	35	15	closed	-1200	yes	27
B5	35	15	closed	-1400	yes	27
B6	35	15	closed	-1600	yes	27
B7	35	15	closed	-1800	yes	27
B8	35	15	closed	-2000	yes	27

Table 4.6: Group B Parameters

In group B simulations with time-varying outlet suction pressure was conducted to characterize air infiltration through the shaft wall gaps. The off gas inlet was maintained at constant flow rate of 35 kg/s and with composition of 60% N₂, 20% CO₂, 5% H₂O, and 15% CO by mass. Air gaps of 27 mm width were defined on all four shaft walls. The outlet suction pressure was ramped from -600 Pa to -2000 Pa to cover the typical operating range of industrial EAF shafts.

B	Pressure (Pa)	Air mass flow (kg/s)	O₂ content (kg/s)	CO mass fraction (-)	CO mass flow (kg/s)	Stoichiometric Ratio
B1	600	11,95	2,74	0,044	2,12	0,91
B2	800	14,54	3,34	0,023	1,11	1,11
B3	1000	17,75	4,08	0,008	0,43	1,36
B4	1200	20,69	4,75	0,003	0,23	1,58
B5	1400	23,17	5,32	0,003	0,22	1,77
B6	1600	25,55	5,87	0	0	1,95
B7	1800	27,68	6,36	0	0	2,12
B8	2000	29,76	6,84	0	0	2,28

Table 4.7 Air infiltration and CO combustion at different suction pressures

Air infiltration and combustion rates at different suction pressures are presented in Table 4.7 and Figure 4.3. The air infiltration increased approximately linearly with suction pressure, ranging from 12 kg/s at -600 Pa to 30 kg/s at -2000 Pa. Oxygen mass flow was calculated as 23% of the total air infiltration.

Stoichiometric ratio = O₂ available / O₂ required for complete CO combustion

The stoichiometric oxygen requirement for complete combustion of the inlet CO (5.25 kg/s at 15% mass fraction) is 3 kg/s O₂. Analysis of Table 4.7 reveals that stoichiometric oxygen availability was achieved at suction pressures as low as -800 Pa. However, CO mass frac-

tion contours at this condition (Figure 4.3) show significant uncombusted CO escaping the shaft despite the presence of adequate bulk oxygen. This observation indicates that CO escape is not limited by total oxygen availability but rather by insufficient mixing between oxygen-rich air streams and CO-rich off gas.

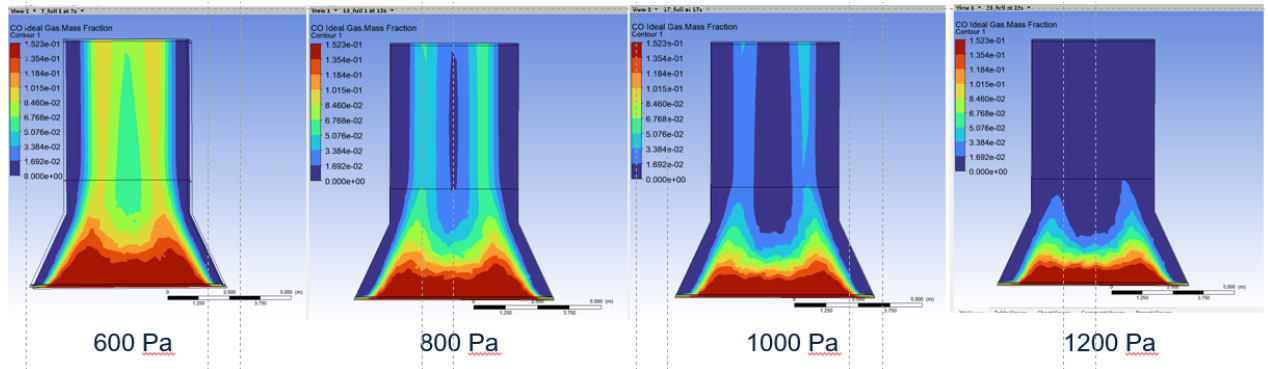


Figure 4.3: CO Concentrations Under Varying Suction Pressure

At -800 Pa, where the stoichiometric ratio exceeded 1.0, the CO contours showed that the CO is leaving the shaft without being fully combusted. Even at -1400 Pa with significant excess oxygen (stoichiometric ratio = 1.77), some amount of CO remained. This diminishing return beyond stoichiometric conditions demonstrates that simply increasing air infiltration does not proportionally improve CO combustion. The limiting factor is the rate at which CO and O₂ can be brought into contact through turbulent mixing, not the absolute quantity of oxygen present in the domain.

In group C, due to the variation of CO concentrations of off gas both in literature and in plant measurements a sensitivity study of CO concentrations was necessary. 8 simulation cases were conducted with increasing CO concentrations starting from 15% to 50% with 5% increments to evaluate the CO combustion behavior with constant false air infiltration. Off gas mass flow rate is kept constant at 35kg/s, constant suction pressure of 1000Pa, scrap is present, and gap width is 27mm.

C	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
C1	35	15	closed	-1000	yes	27
C2	35	20	closed	-1000	yes	27
C3	35	25	closed	-1000	yes	27
C4	35	30	closed	-1000	yes	27
C5	35	35	closed	-1000	yes	27
C6	35	40	closed	-1000	yes	27
C7	35	45	closed	-1000	yes	27
C8	35	50	closed	-1000	yes	27

Table 4.8: Group C Parameters

Simulation results of given 8 cases show that at 15% CO concentration, approximately 95% of the CO is combusted. Whereas when the CO concentration is increased to 50% only 60% of present CO combustion is achieved. Based on the results from other simulations and plant experiments it is known that the air flow into the shaft can be increased either by increasing the suction pressure or the gap width but both in real plant experiments and CFD simulations it is seen that CO and O₂ still can be found at the shaft outlet which indicates that the mixing is a greater limitation than oxygen deficiency.

C	O2 Mass Flow (kg/s)	CO Mass Fraction	CO Mass Flow (kg/s)	CO2 Mass Fraction	False Air (kg/s)	Combusted CO%
C1	1,42	0,0086	0,46	0,27	18	94,26
C2	0,87	0,022	1,21	0,3	18	89
C3	0,47	0,04	2,27	0,32	18	84
C4	0,22	0,06	3,62	0,33	18	80
C5	0,09	0,09	5,21	0,34	18	74,28
C6	0,033	0,12	6,9	0,34	18	70
C7	0,0079	0,16	8,63	0,34	18	64,44
C8	0,0008	0,19	10,34	0,34	18	62

Table 4.9: Group C Results

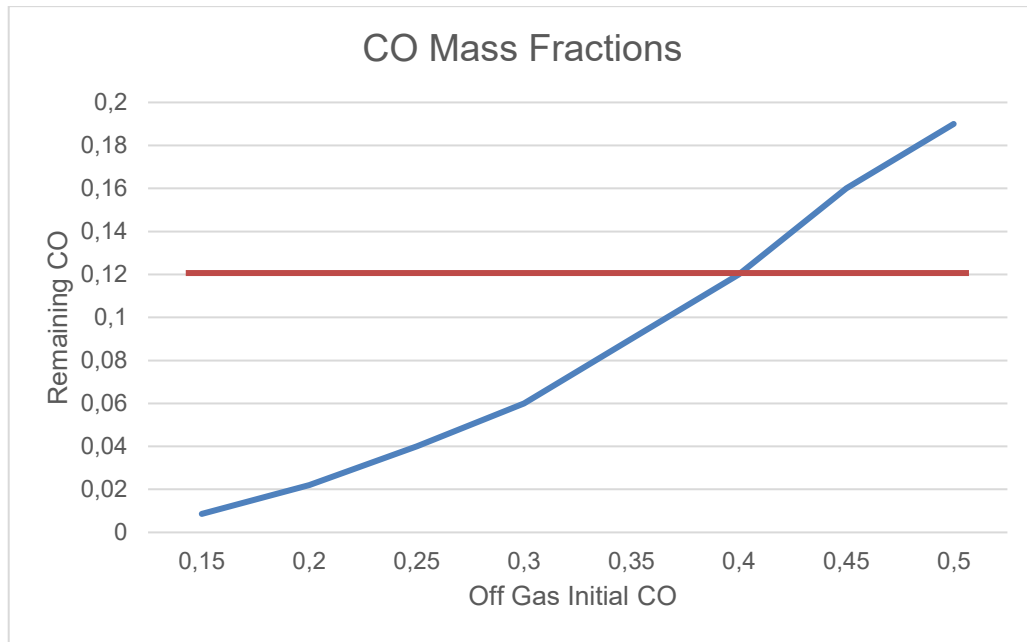


Figure 4.4 CO Mass Fractions

Figure 4.4 shows the remaining CO mass fractions after combustion for cases with increasing initial CO mass fractions from 15% to 50%. It is considered that CO mass fractions over 12% in off gas is dangerous since it can cause explosions down the suction line. Under normal operating conditions with a negative pressure of -1000Pa inside the furnace and gap width of 27mm, carbon and oxygen addition might need to be optimized in a way to keep off gas CO mass fraction under 40% inside the furnace.

In group D all the parameters such as mass flow rate, CO concentrations, suction pressure were kept the same, flap is closed and scrap is present. The width of air infiltration gaps are increased 2, 3 and 4 times as it is often applied at steel plants. At a suction pressure of 1000Pa which is the usual suction pressure during operation an increase from approximately 16 to ~23 kg/s is observed when the gap width increased to 54mm from 27mm. Further increase slightly caused more false air flow into the system. At 81mm gap the false air flow rate increased to 26kg/s and at 108mm it increased to 27kg/s showing diminishing returns after 54mm.

D	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
D1	35	15	closed	-1000	yes	27
D2	35	15	closed	-1000	yes	54
D3	35	15	closed	-1000	yes	81
D4	35	15	closed	-1000	yes	108

Table 4.10: Group D Parameters

In this group the effect of increased false air flow by increased gap width on CO combustion and scrap preheating was observed. Simulations show that by doubling the gap width the mass fraction of CO drops from 13% to 0,5% and it doesn't significantly change by further increasing the gap width.

D	O2 mass fraction	O2 mass flow (kg/s)	CO mass fraction	CO mass flow (kg/s)	CO2 mass fraction	False Air (kg/s)
D1	0,002	1,14	0,133	0,687	0,27	16,56
D2	0,041	2,43	0,005	0,29	0,25	23,37
D3	0,049	3	0,004	0,27	0,25	26
D4	0,05	3,24	0,004	0,25	0,24	27

Table 4.11: Group D Results

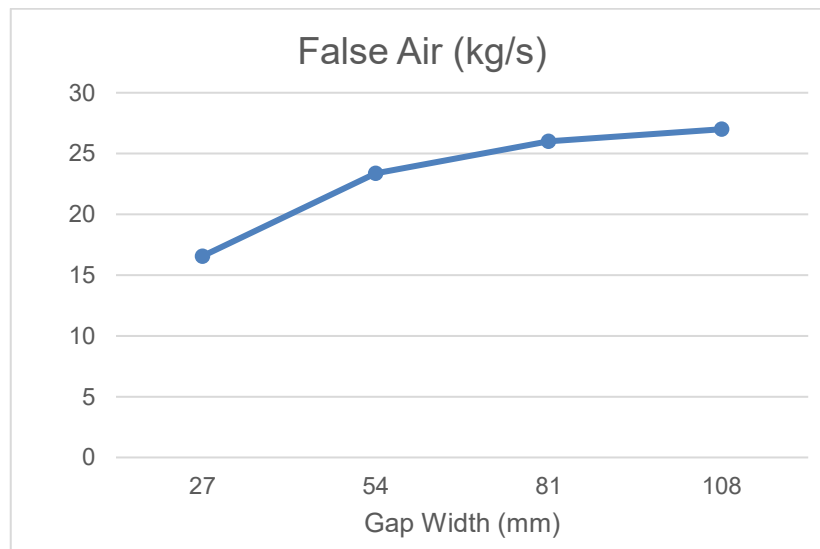


Figure 4.5 Increase of False Air with Gap Width

Figure 4.5 shows the relationship between the gap width and the false air flow.

In group E off gas mass flow rate and gap widths were kept the same, scrap is present, and the flap is closed. In this group of cases the aim is to understand the effect of the CO combustion on gas and scrap temperatures.

E	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
E1	35	15	closed	-500	yes	54
E2	35	15	closed	-1000	yes	54
E3	35	15	closed	-1200	yes	54
E4	35	15	closed	-2000	yes	54

Table 4.12: Group E Parameters

In 4 cases the suction pressure is increased from 500Pa to 1000, 1200 and 2000Pa. Simulation results showed that the CO mass fraction is lowered to 12% with 1.1kg/s false air mass flow rate at 500Pa suction pressure and to less than 1% with 28kg/s false air mass flow rate at 2000Pa suction pressure.

E	Average Scrap Tem- perature (K)	Maximum Scrap Tem- perature (K)	Average Gas Temperature (K)	Maximum Gas Tempe- rature (K)	O2 mass fraction
E1	930	1402	1263	2047	0
E2	1048	1440	1520	2186	0,01
E3	1049	1422	1524	2162	0,02
E4	1026	1413	1464	2108	0,05
	O2 mass flow (kg/s)	CO mass fraction	CO mass flow (kg/s)	CO2 mass fraction	False Air (kg/s)
E1	0	0,12	4,5	0,21	1,1
E2	0,58	0,04	1,85	0,26	10,95
E3	1,15	0,02	1,36	0,26	14,8
E4	3,6	0,005	0,32	0,23	28

Table 4.13: Group E Results

With increasing CO combustion rate the scrap temperatures has increased at 1000 and 1200Pa suction pressures.

As expected, higher scrap temperatures are observed in simulation results due to increase in false air flow rate. Volume averaged scrap temperature of 930K is obtained at the lowest false air mass flow rate case of 1.1kg/s and it increased to approximately 1050K with the mass flow rates of 11 and 15kg/s. Further increase of suction pressure beyond 1200Pa

shows diminishing returns at scrap temperatures being 1026K at 2000Pa suction pressure with a false air flow rate of 28kg/s. This shows that even if it is possible to effectively combust CO by increasing the suction pressure, it may also diminish the scrap preheating performance.

However, the exact value of suction pressure where the diminishing returns start is difficult to assess since the off gas CO concentration and mass flow rate can vary from operation to operation and throughout a single operation.

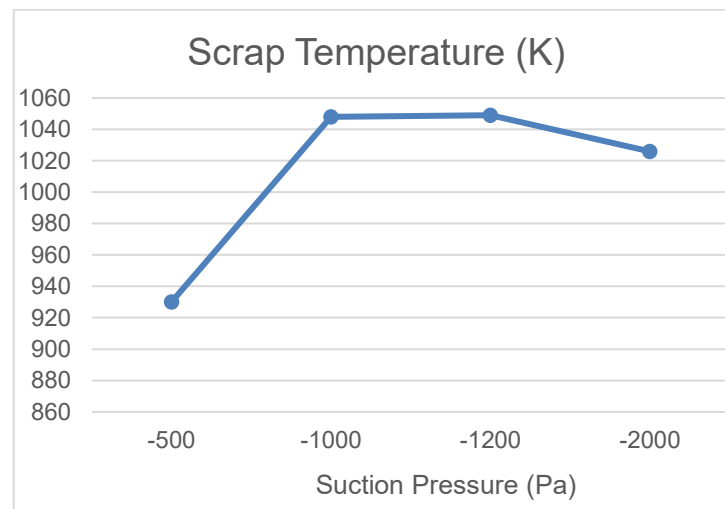


Figure 4.6 Scrap Temperatures at Different Suction Pressures

Figure 4.6 shows the relationship between the increased suction pressure and the scrap temperatures. Increasing the suction pressure has a stronger effect on the false air flow. Doubling the suction pressure from 1000Pa to 2000Pa almost tripled the suction flow

F	Mass Flow (kg/s)	CO %	Flap	Pressure (Pa)	Scrap	Gap (mm)
F1	35	15	closed	-1000	yes	27-all sides
F2	35	15	closed	-1000	yes	27- front/back
F3	35	15	closed	-1000	yes	27-only sides

Table 4.14: Group F Parameters

In almost all simulation cases some amount of uncombusted CO leaving the shaft is observed and this finding is aligned with reported problems from the plants. Simulations and real plant experiments of increasing false air flow by either increasing gap widths or suction pressures still result in a low amount of CO escaping the shaft and causing explosion further through the suction line. Simulations show that the temperatures are suitable for CO combustion, and the required oxygen is present, but the CO still escapes. This rules out the oxygen deficiency and temperature limitations from being the main reason for this CO runaway.

This leaves the mixing limitation of the off gas and false air as the most likely cause. To understand how the off gas and false air interact, the gaps that were defined around the shaft are divided into groups. The first group includes all the gaps at 4 sides of the shaft which is the default arrangement. The second group includes the gaps at the front and back faces of the shaft, and the third group includes the gaps at the left and right sides of the shaft.

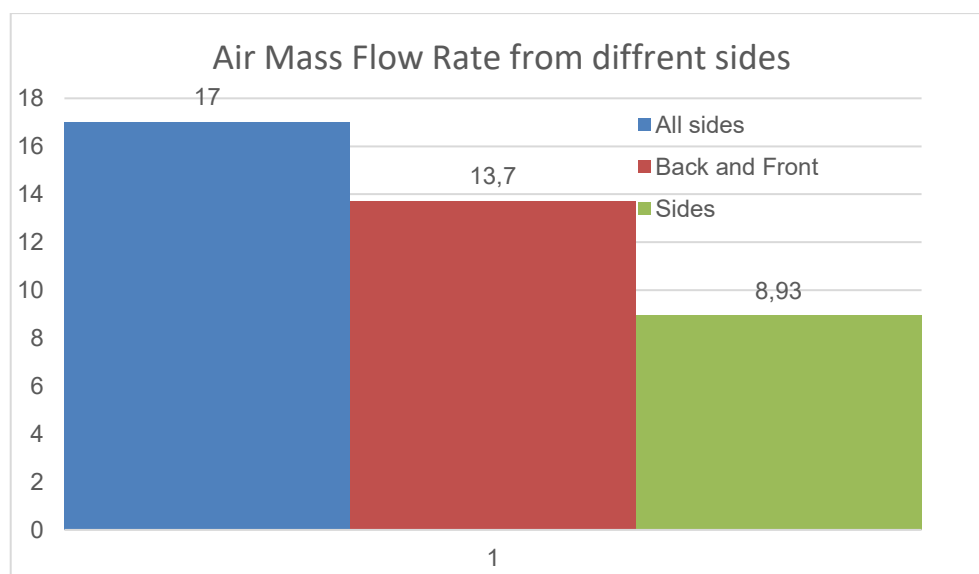


Figure 4.7: Group F False Air Mass Flow Rates

The simulation results of this three groups show that most of the false air infiltrates the system through the gaps at the front and back side. This result is expected since the front and back faces of the shaft have a larger width which makes the gaps at the front and back side also larger compared to the sides. When the air infiltration is allowed only from two sides whether it's the front and back or left and right, the individual false air flow rate from those gaps increase compared to the case with false air infiltrating from all sides. This causes the sum of flow rates of separate cases of different sides to have greater flow rate than the case with all sides open.

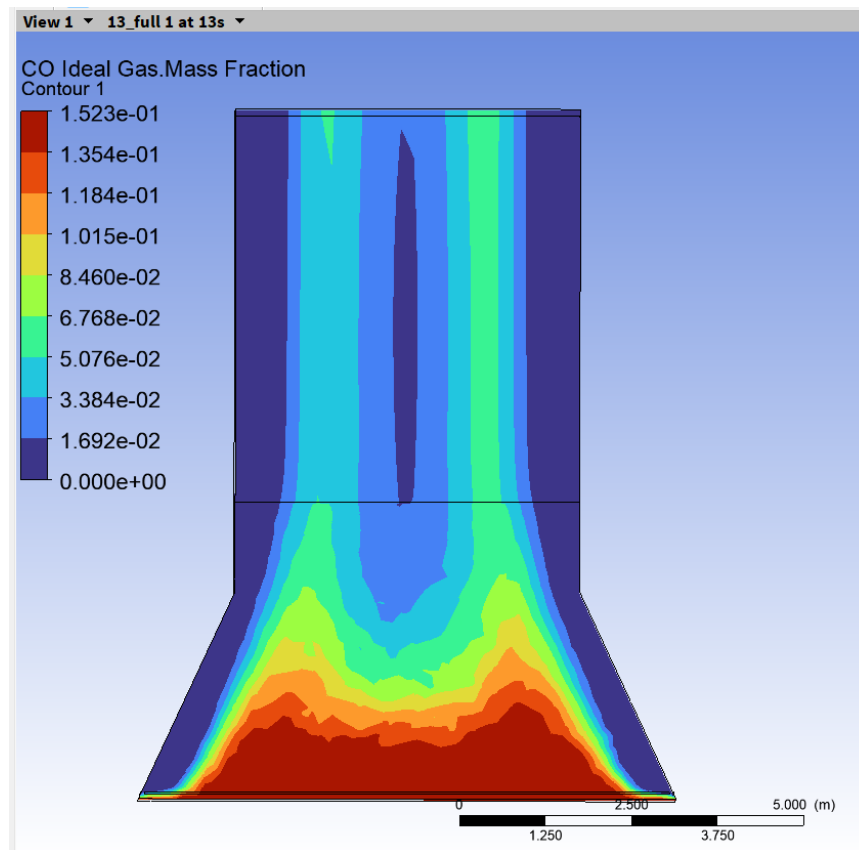


Figure 4.8: Case F1 CO Mass Fraction Contour with Air Infiltration From All Sides

In the case where the false air can infiltrate the system from all sides a pattern of two streamlines that are rich in CO leaving the shaft is observed. This pattern is consistent throughout all the cases previously mentioned where CO is not fully combusted. This indicates that CO and O₂ are leaving the shaft separately and to understand how, CO mass fraction contours are also observed in remaining cases of separate gaps.

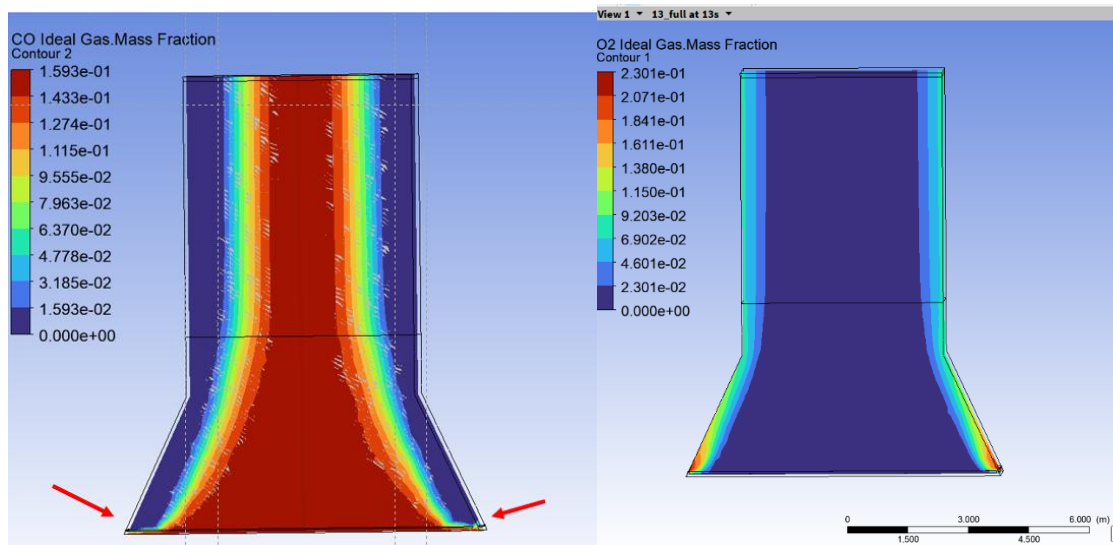


Figure 4.9: Case F2 O₂ and CO Contours with Air Infiltration From Left and Right

The case where the false air flow is only allowed from the sides shows that the air coming in from the sides is not penetrating deeply into the shaft and follows the wall length throughout the shaft. Some reaction takes place where CO and air streams come in contact due to diffusion, but the flowing air doesn't cause any significant entrainment or mixing. Even though the scrap would affect the gas flow inside the shaft, this behavior is expected since the scrap porosity reaches to 1 at the shaft walls. So, the air can flow through the sides whereas the off gas can find its path through the scrap.

In contrast to that when air infiltration is allowed only from the front and back sides, lower CO concentrations at the middle and high CO concentrations on the right and left side of the shaft is observed.

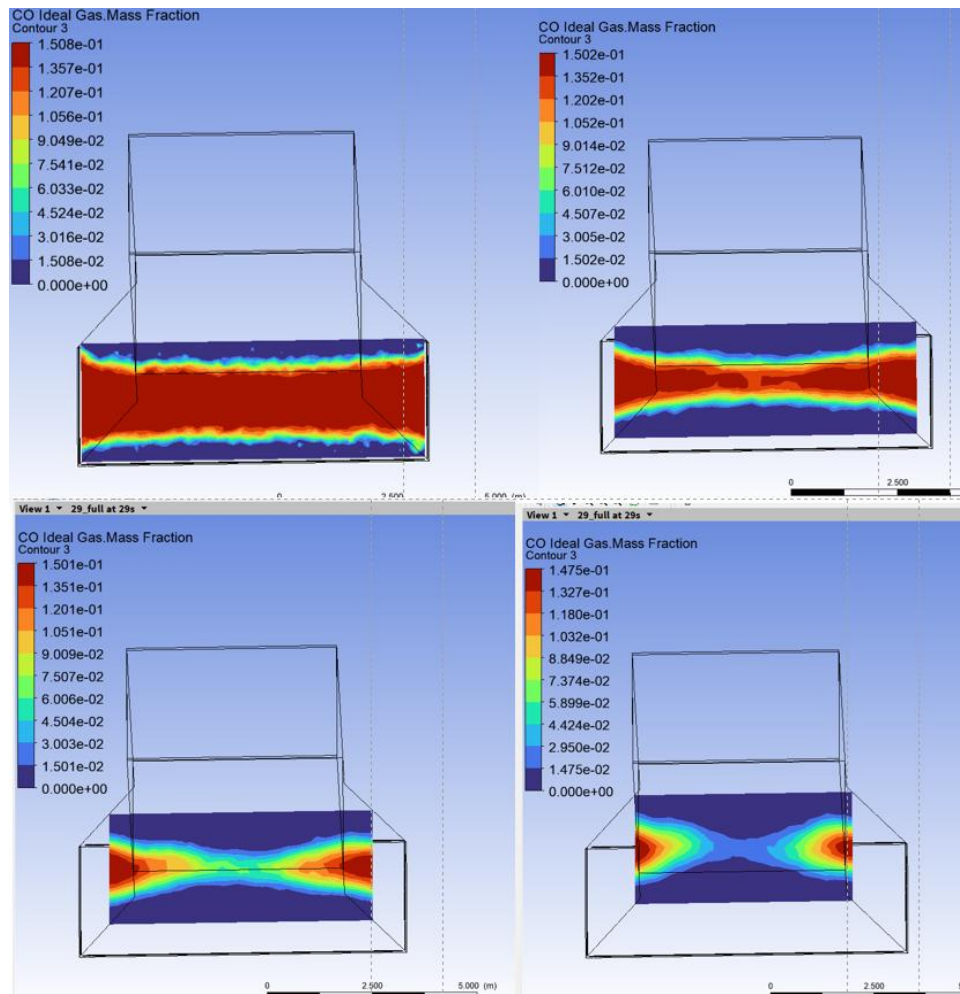


Figure 4.10 Case F3 CO Contour XY Section Plane with Air Infiltration from Back and Front

The false air leaking from the front and back sides can penetrate all the way to the middle of the shaft only leaving streams of high CO concentration off gas adjacent to the walls due to the shaft geometry.

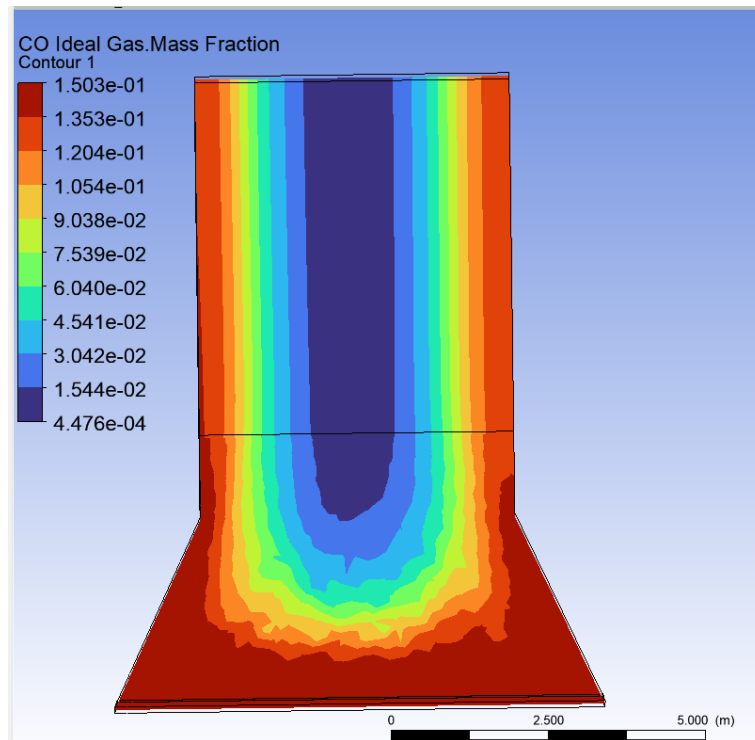


Figure 4.11: Case F3 CO Contour XZ Section Plane with Air Infiltration from Back and Front

CO mass fraction profiles extracted along horizontal lines at the outlet plane show characteristic double-peak distributions. For the baseline configuration, local CO mass fractions reach 0.10-0.15 at the peak locations, while the center and near-wall regions show values below 0.03. This indicates that these are not random fluctuations but geometrically determined features of the flow structure.

The persistence of these CO-rich zones even when bulk oxygen availability exceeds stoichiometric requirements demonstrates that their formation is not due to insufficient oxidizer. Instead, the pattern shows the three-dimensional flow structure created by the interaction of multiple air infiltration streams within the porous scrap bed geometry. Regions that are geometrically isolated from the primary air entry points receive oxygen only through slower diffusive transport, while the high-velocity air streams create preferential combustion zones where direct contact occurs.

Discussion

Heat Transfer Performance and its Importance

The baseline scrap preheating simulations show that convective heat transfer between the off-gas and the packed scrap bed is strongly governed by both the gas mass flow rate and the particle diameter. Convective heat transfer coefficients calculated through the Gunn correlation ranged from approximately $36 \text{ W}/(\text{m}^2 \cdot \text{K})$ at a mass flow rate of 5 kg/s to approximately $119 \text{ W}/(\text{m}^2 \cdot \text{K})$ at 45 kg/s for the baseline particle diameter of 0.1 m . These values are physically consistent with the high porosity of the bed ($\epsilon = 0.85$) and the turbulent flow regime confirmed by particle Reynolds numbers in the range of approximately 500 to 4300.

The parametric study across particle diameters from 0.05 m to 0.30 m demonstrates a clear and physically expected inverse relationship between particle size and heat transfer coefficient. At 45 kg/s off-gas flow, reducing the particle diameter from 0.30 m to 0.10 m roughly doubles the convective heat transfer coefficient, from approximately $67 \text{ W}/(\text{m}^2 \cdot \text{K})$ to $119 \text{ W}/(\text{m}^2 \cdot \text{K})$. This behavior follows the Gunn correlation and the interfacial area density scaling as $6(1 - \epsilon)/d_p$, which increases as particle diameter decreases. The practical consequence is that scrap pieces with high surface area to volume ratios, such as sheet metal, rebars, and shredded material, heat substantially faster than large structural pieces or heavy rails despite experiencing the same bulk gas temperature and the same superficial velocity.

This finding provides an explanation for plant observations of localized scrap melting and dripping at the base of the shaft during preheating, which would appear inconsistent with the moderate predicted bulk temperatures of approximately 325°C after five minutes under baseline conditions. In a real mixed scrap batch, thin sections with characteristic dimensions on the order of a few millimeters can experience heat transfer coefficients far greater than those calculated here for representative bulk particles, and can therefore reach temperatures approaching the melting temperature of steel locally, while the bulk of the heavier charge remains below 600 K . The CFD model cannot capture this heating behavior within the uniform porous medium assumption, which must be recognized as a structural limitation of the homogeneous porous medium assumption rather than a numerical error.

Vertical shaft preheating systems of the Quantum type are reported in the literature to achieve energy savings of maximum 100 kWh/t-steel under ideal conditions, with maxi-

imum temperatures approaching 773 K for the scrap bulk [Lee and Sohn, 2014]. The present simulation results for average scrap temperatures after 450 seconds of preheating at the baseline mass flow rate of 35 kg/s show a volume-averaged scrap temperature of approximately 1050 K, which corresponds to a sensible heat uptake per ton of scrap of approximately 104 kWh/t. This is in reasonable agreement with the expected energy recovery range for this system.

$$Q = m_{steel} \times c_{p,steel} \times (T_{final} - T_{initial})$$

$$Q_{@1000,1200Pa} = 1000kg \times 0,49 \text{ kJ}/(kg \cdot K) \times (1050K - 300K)$$

$$Q_{@1000} \approx 104kWh/t$$

Suction Pressure (Pa)	Q (kWh/t)
500	$\approx 87,5$
1000,1200	≈ 104
2000	≈ 100

Table 4.15: Total Energy to Scrap at Different Suction Pressures

Table 4.15 shows the heat absorbed per ton of scrap. The simulation results are comparable to plant data for energy consumption per ton of steel at different suction pressures. The data received from the plant is shown in the figure 4.12. Plant data shows that the energy usage drops by 20kWh/t when the suction pressure is increased to 1000Pa from 500Pa (plant values are measured in furnace as mbars showing negative values. -5mbar furnace pressure corresponding to 500Pa suction pressure from the PCC.)

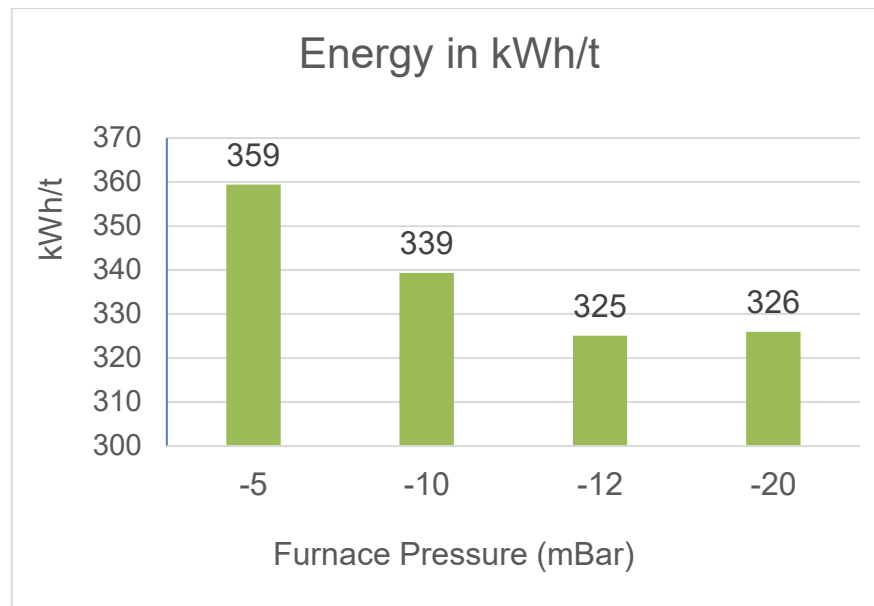


Figure 4.12 Energy Consumption Under Different Pressures

Figure 4.12 shows the energy consumption in kWh per ton of steel from an actual steel plant with corresponding suction pressures. The energy consumption is decreasing as the suction pressure increases from 500Pa to 1200Pa and stagnates or slightly increases after further increase in the suction pressure.

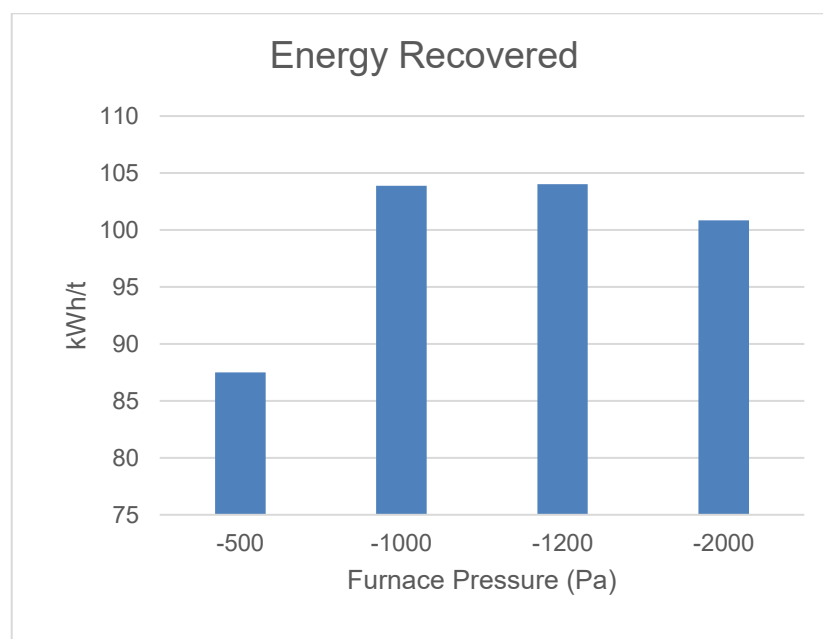


Figure 4.13 Energy Recovered to Scrap

As it is shown in the Table 4.7 for group B, 600Pa of suction pressure is not enough to supply enough oxygen through false air into the system. Therefore, in Figure 4.12 the energy consumption decreases as the pressure goes from -5 to -10 and -12.

Simulation results and the plant data is in alignment on tradeoff between the scrap preheating performance and increasing false air flow rate for better CO combustion. Knowing that increased false air flow does not have a linear correlation with CO conversion rate it can be suggested that instead of relying on false air flow into the system driven by the suction pressure, controlled air, pure oxygen or a mixture would yield better results. By injecting the oxidizer with a higher velocity into the shaft, better mixing can be achieved therefore all the CO can be theoretically combusted without sucking excess amounts of air into the shaft.

Higher scrap temperatures and CO conversion can be achieved by injecting the oxidizer into the shaft, but it requires further evaluation. Turbulent kinetic energy across the shaft needs to be determined and correlated with local CO conversion rates to identify the mixing-deficient zones where incomplete CO combustion persists. Targeted injection of oxidizer into these specific zones at sufficiently high velocity would generate turbulent shear and entrainment of the surrounding gas and improve the local mixing intensity without the thermal penalty associated with flooding the shaft with excess false air. A controlled oxidizer injection offers the additional advantage of decoupling the combustion optimization from the suction pressure boundary condition, allowing independent control over both the quantity and spatial distribution of the oxidizer. Therefore, further parametric study is required to evaluate the optimal injection velocity, nozzle positioning, and oxidizer composition that maximizes CO conversion while preserving the scrap preheating performance of the shaft system.

Mixing Limited CO Combustion and Explosions

The main finding of the combustion investigation is the demonstration that incomplete CO combustion in the EAF shaft is not caused by the deficiency of available oxygen under normal operating conditions, but by insufficient turbulent mixing between the air infiltration streams and the CO-rich off-gas. This conclusion is supported by converging evidence from three independent groups of simulation analysis: the stoichiometric ratio study

(Group B), the CO concentration sensitivity study (Group C), and the gap width variation study (Group D).

Group B simulations quantified the relationship between suction pressure and the air infiltration rate through the 27 mm gap configuration. The stoichiometric oxygen requirement for complete combustion of the inlet CO at 35 kg/s and 15 % mass fraction is approximately 3 kg/s of O₂, corresponding to approximately 13 kg/s of air. The results show that this stoichiometric threshold is already exceeded at a suction pressure of only 800 Pa, at which point the air mass flow rate reaches approximately 14.5 kg/s and the O₂ supply corresponds to a stoichiometric ratio of 1.11. Despite this, the CO mass fraction at the shaft outlet at this condition remains above zero (approximately 0.023 from Table 4.7), demonstrating that the availability of sufficient bulk oxygen is not the rate-limiting step. This observation contradicts the assumption made in several existing EAF process models, in which all oxygen entering through air infiltration is assumed to react completely with CO regardless of mixing conditions. Bekker et al., as described in Hay et al. [2021], represent this class of model in which CO post-combustion rate is effectively limited only by air leakage rate rather than by mixing quality.

The persistence of uncombusted CO is not attributable to thermal quenching either. The inlet gas temperature of 1500 K is far above the CO autoignition temperature, and the Damköhler number analysis confirmed that the chemical timescale of approximately 7.67×10^{-5} s is negligible compared to the gas residence time of approximately 0.92 s at 35 kg/s. The resulting Damköhler number of approximately 12,000 at this flow rate confirms that any CO molecule that comes into contact with O₂ at these temperatures will react completely within a timescale orders of magnitude shorter than its residence time. Therefore, the presence of unreacted CO at the outlet indicates that not all CO molecules encounter adequate oxygen concentrations during their transit through the bed. The limiting step is the rate at which turbulent eddies transport the reactants into contact, not the rate at which they react once in contact.

The Group C results reinforce this interpretation through a different approach. Increasing the inlet CO mass fraction from 15 % to 50 % while maintaining constant air infiltration conditions (suction pressure 1000 Pa, gap width 27 mm, mass flow rate 35 kg/s) produces a progressively declining CO combustion efficiency, from approximately 94 % at 15 % CO to approximately 62 % at 50 % CO (Table 4.9). If the process were genuinely mixing-

limited in the geometric sense that certain zones receive insufficient oxygen regardless of the bulk supply, then increasing the fuel load in those zones while holding the air distribution constant should reduce combustion efficiency in direct proportion to the fuel enrichment. The fact that even at 15 % CO inlet concentration, approximately 6 % of the CO escapes uncombusted despite an overall air supply exceeding stoichiometric requirements, demonstrates that the spatial separation of reactant streams is a persistent geometric feature of the system, not a transient deficiency that can be resolved by adjusting flow rates.

These findings provide a possible explanation for the explosions observed in the steel plants. Previous experiments were conducted in plant to overcome this problem of explosions by flooding the shaft with high amounts of false air during the carbon charging into the furnace and despite the large amounts of false air infiltrations, explosions took place shortly after the carbon addition. When the added CO content of the off gas increases as well as the oxygen content from the false air but since they leave the shaft separately without reacting, the CO cools down after passing through the scrap, then they get mixed down the suction line. When this cooled CO and O₂ mixture finds required temperatures for combustion in the post combustion chamber they cause strong explosions.

Effect of Geometry on Air-CO Interaction: Group F Analysis

The Group F simulations, in which air infiltration was restricted to specific faces of the shaft geometry, provide the most direct insight into the three-dimensional flow structure that determines CO combustion efficiency. By isolating the contributions of front/back infiltration from side infiltration, these cases reveal that the two gap configurations serve complementary but geometrically constrained roles within the shaft.

When air infiltration is permitted only through the front and back faces (Case F2), the results show that oxygen-bearing air streams penetrate deeply into the cross-section of the shaft, generating a combustion zone in the central region of the bed. However, this configuration leaves the regions adjacent to the side walls oxygen-depleted, producing elevated CO mass fractions along the left and right boundaries of the shaft at the outlet plane. This behavior is consistent with the trapezoidal cross-section of the Quantum shaft geometry, where the front and back faces have greater width and therefore admit more air per unit of suction pressure, allowing their streams to converge toward the geometric center.

Conversely, when air infiltration is restricted to the side faces only (Case F3), the infiltrating air follows the near-wall path of low flow resistance created by the elevated porosity approaching unity at the solid boundaries. Rather than penetrating the packed bed, the side air streams flow upward along the walls, providing oxygen near the walls but fails to mix with the high-CO off-gas that occupies the center of the shaft. The CO mass fraction contours in Figure 4.11 show that the central region of the shaft at the outlet plane remains CO-rich under this configuration, while the near-wall zones show lower CO concentrations where air and off-gas have come into contact through diffusion across the shear layer between the fast-moving wall stream and the slower core flow.

The default four-sided configuration (Case F1) represents an overlap of these two behaviors, with neither the center nor the walls mixing fully. The consistent double-peak structure observed in the CO mass fraction profiles at the outlet plane, with elevated CO concentrations at two symmetric positions, shows the geometric constraints of the shaft: front/back air creates a combustion zone in the center but pushes CO toward the sides, while side air creates a near-wall combustion zone but leaves the center incompletely combusted. These two CO rich regions correspond to the locations that are geometrically furthest from the air entry points for each respective infiltration direction.

Another observation is that the near-wall porosity effect cannot be corrected by simply increasing the total air flow rate. Because the elevated porosity at the walls is a geometric consequence of the scrap packing arrangement rather than a controllable parameter, side-entry air will always follow the wall path regardless of the infiltration rate. This means that the mixing limitation near the sidewalls cannot be resolved by operating adjustments alone. Targeted oxidizer injection, offers a possible solution to overcome this constraint.

Artificial Oxygen Supplementation

The combined findings of this investigation suggest a design requirement for improving CO post combustion performance in EAF shaft preheating systems. The solution is not to increase the rate of air infiltration, but targeted delivery of either oxygen or air into the zones where CO accumulates and mixing is limited. The results identify two primary regions that require attention: the sidewall zones in the upper portion of the shaft, where side-entry air fails to penetrate the packed bed core, and the lateral regions near the front and back walls in the lower shaft, where CO-rich streams from the off-gas inlet avoid the central combustion zone created by front/back air.

Targeted injection at these specific locations, would introduce oxygen at points where it is currently absent and would do so with sufficient momentum to generate turbulent shear and entrainment of the surrounding CO-rich gas. The Damköhler number analysis confirms that once CO and O₂ are brought into contact at these temperatures, complete reaction will follow on a chemical timescale many orders of magnitude shorter than the flow timescale. The combustion efficiency achievable by targeted injection is therefore limited only by the spatial coverage and mixing intensity of the injection, not by any kinetic barrier.

Additional simulations were conducted to better understand the concept of artificial oxygen supplementation. Injection holes that are targeting the streamlines where CO is escaping are defined and based on the stoichiometric requirements, oxygen is supplied. For the case where air infiltration was only allowed from back and the front sides of the shaft, the holes are modeled at the inclined side walls of shaft.

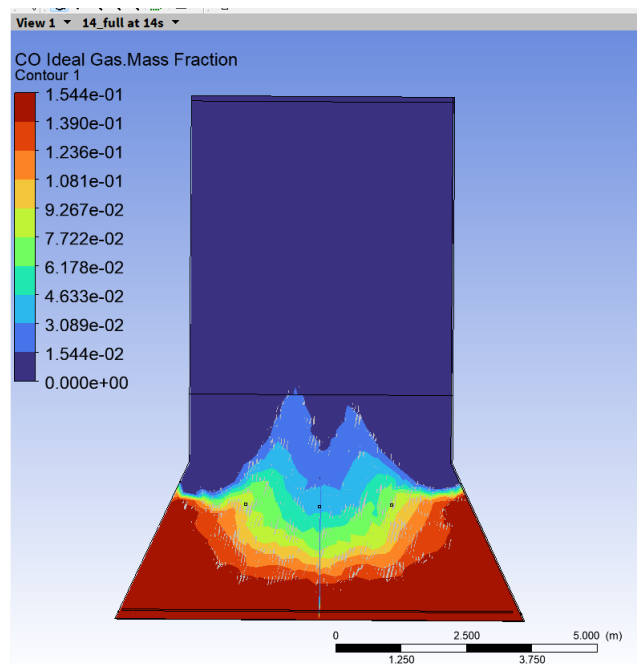


Figure 4.14 CO Combustion with Artificial Oxidizer Supplementation

In the CO mass fraction contour from figure 4.14 it is seen that the complete CO combustion is achieved by the help of additional burners targeting the trapped CO near the side walls.

The advantages of this approach over increased suction pressure or larger gap widths are the decoupling of combustion optimization from the thermal penalty of excess air and controllable flow rates. Stoichiometric or near-stoichiometric injection of pure oxygen into the CO accumulation zones would supply the necessary oxidizer without introducing large

masses of inert nitrogen that extract heat from the scrap. Based on the stoichiometric analysis in Section 3, the oxygen requirement for complete combustion of the inlet CO at 35 kg/s and 15 % mass fraction is approximately 3 kg/s. This is a small flow rate relative to the total gas throughput, and its injection at certain locations would represent a minor addition to the overall flow field while providing the locally intense mixing needed to achieve complete combustion. Quantitative optimization of injection velocity, nozzle position, and oxidizer composition (air versus oxygen-enriched air versus pure oxygen) requires dedicated parametric CFD studies beyond the scope of the present work but is supported by this study.

Validation Against Plant Observations and Model Consistency

Although direct validation of CFD predictions against detailed measurements inside an operating EAF shaft is not feasible in EAF conditions, several indirect comparisons with available plant data and literature values confirm that the model captures the essential physics of the system.

A consistent finding across Groups B, D, E, and F is that increasing the rate of cold air infiltration into the shaft produces competing effects on system performance: combustion efficiency improves as more oxygen becomes available and greater turbulent shear promotes mixing, but at high enough infiltration rates scrap preheating performance begins to degrade because the excess nitrogen and unused oxygen carry heat out of the shaft rather than depositing it in the scrap. Critically, the new simulation results from Group E demonstrate that this trade-off is not a simple monotonic penalty but rather exhibits an optimum, and this non-monotonic behavior is in quantitative agreement with the plant energy data presented in Figure 4.12.

The qualitative agreement between the simulated pressure-concentration relationships in Figure 3.5 and the plant measurements in Figure 2.17 provides additional confidence in the model boundary conditions. Both data sets show that CO and CO₂ concentrations increase when furnace pressure rises and false air infiltration decreases, and that CO concentrations drop when the flap opens and the chimney effect draws large air volumes into the shaft. The simulation results for Group A cases with the open flap (A1-A3) correctly predict that CO-rich off-gas leaves the system rather than false air entering, consistent with the plant observations of flames outside the shaft and at the hood during the charging phase.

The Biot number analysis confirmed that all particle sizes considered in parametric study satisfy the lumped capacitance criterion ($Bi < 0.1$ for all cases), validating the use of a single solid-phase temperature in the energy equation without internal conduction modeling. The mesh independence study demonstrated that key temperature predictions change by less than 2 % between the medium and fine meshes, providing confidence that discretization errors are not contributing significantly to the results. Together, these verification and validation checks support the conclusion that the simulation results, while subject to the structural limitations of the homogeneous porous medium assumption, provide a physically consistent and reliable basis for the design recommendations developed in this chapter.

Model Assumptions and Limitations

Several simplifying assumptions in the CFD model limit the quantitative accuracy of absolute predictions, though relative comparisons between cases remain valid for optimization purposes. The most significant assumption is the representation of scrap as a homogeneous

porous medium with uniform spherical particles. Real scrap comprises a heterogeneous mixture of large chunks, sheet metal, and irregular shapes with size distribution spanning two orders of magnitude. This heterogeneity creates preferential flow paths (channeling) and local variations in porosity that are not captured in the uniform porous model. The particle size sensitivity study in Section 5.1.2 demonstrates that this assumption affects predicted temperature distributions and heat transfer rates.

Chemical kinetics were represented using a simplified global mechanism (Westbrook-Dryer) or the Eddy Dissipation Model. Both approaches neglect detailed CO oxidation chemistry including radical formation and multi-step reaction pathways. However, the demonstration that oxygen and CO coexist in the mixing-limited zones indicates that turbulent mixing rather than chemical kinetics controls the overall reaction rate, reducing the importance of detailed kinetic modeling for the present optimization objectives.

Wall heat losses were estimated based on typical furnace shell temperatures and external convection correlations. Actual heat loss distributions depend on refractory condition, external insulation that were not modeled in detail. This affects the predicted scrap temperatures but has minimal impact on the gas-phase mixing and combustion behavior that is the focus of the optimization study.

Despite these limitations, the model successfully captures the essential physics of air infiltration, CO-O₂ mixing, and combustion in the geometry of an EAF scrap preheating shaft. The comparison of multiple configurations under consistent modeling assumptions provides reliable guidance for design optimization, and the identified mechanisms (mixing limitation, complementary roles of different air sources) are robust to the model uncertainties.

5. Summary

This thesis investigated CO combustion and scrap preheating in EAF shaft furnaces using CFD. The motivation was a recurring problem observed in industrial plants: CO escapes the shaft uncombusted despite high gas temperatures and the presence of oxygen, causing energy losses, safety risks from downstream ignition, and unnecessary carbon emissions.

The fundamental reason for this is not known, and direct measurement inside an operating shaft is not feasible.

A CFD model was built in ANSYS to simulate the coupled problem of heat transfer, gas flow, and combustion inside the shaft. The scrap bed was represented as a porous medium using volume-averaged conservation equations, with flow resistance from the Ergun equation and gas-solid heat transfer from the Gunn correlation. Thermal non-equilibrium between the gas and solid phases was captured through separate energy equations. CO combustion was modeled with the Eddy Dissipation Model, which was validated as appropriate through Damköhler number analysis confirming that reaction rates are much faster than mixing rates. Temperature-dependent gas properties were calculated using VDI Heat Atlas correlations. A mesh independence study confirmed that the medium mesh with approximately 500,000 elements provides sufficient accuracy.

Baseline preheating simulations without combustion showed that heat transfer performance is strongly dependent on both mass flow rate and particle size. Convective heat transfer coefficients increase from around 36 to 119 W/(m²·K) as mass flow rate increases from 5 to 45 kg/s. Smaller particles heat substantially faster due to their higher surface area per unit volume, which explains why thin sheet metal and small scrap pieces can reach melting temperature locally while the bulk scrap temperature remains moderate. This finding is consistent with plant observations of localized dripping and melting that bulk temperature predictions alone cannot explain.

Combustion simulations demonstrated that stoichiometric oxygen from air infiltration is already available at suction pressures as low as 800 Pa under normal operating conditions. However, CO was still found escaping the shaft at this and higher pressure levels. This result rules out oxygen deficiency as the primary cause of incomplete combustion and establishes that the limiting factor is mixing. CO and air enter the shaft from different locations and do not come into sufficient contact before reaching the outlet.

Parametric studies on CO concentration showed that combustion efficiency decreases progressively as CO concentration increases from 15% to 50%, while oxygen supply remains constant. Increasing gap width and suction pressure both increase air infiltration, which improves combustion but reduces scrap temperatures after a certain point because the ex-

cess cold air absorbs heat that would otherwise go into the scrap. These two effects cannot be decoupled when using bulk air infiltration as the only control variable.

Isolated infiltration configuration tests revealed that front and back gaps drive air into the center of the shaft, creating a combustion zone there but leaving the sidewall regions with high CO concentrations. Side gaps supply oxygen near the walls but the air flows along the wall without deeply penetrating the packed bed, so the center region remains oxygen-deficient when only side infiltration is active. Both infiltration directions serve a purpose, but neither alone is sufficient for complete combustion. The combined default configuration produces a consistent two-peak CO escape pattern at the outlet that is geometrically determined by the shaft and gap geometry rather than by operating conditions.

These findings indicate that the CO escape problem in EAF shafts is a mixing problem. The solution is not more air, but better-placed oxidizer. Targeted injection of oxygen or air directly into the identified CO accumulation zones, particularly near the sidewalls in the upper section of the shaft, offers the potential to significantly improve CO conversion without the thermal penalty associated with flooding the shaft with excess cold air. This decouples combustion optimization from the suction pressure boundary condition and allows independent control over where oxidizer is introduced.

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